A T M E College of Engineering

Subject Name: Microwave Engineering & Antenna Theory

Subject Code: BEC701

Course Coordinator:

Guruprasad K N

Assistant Professor

Dept. of E&C, ATME

ATME College of Engineering

Module 5

Loop and Horn Antenna
&
Antenna Types

Module 5: Chapter 1

Loop and Horn Antenna:

- Introduction
- Small loop
- Comparison of Far fields of Small Loop and Short Dipole
- The Loop Antenna General Case
- Far field Patterns of Circular Loop Antenna with Uniform Current
- Radiation Resistance of Loops
- Directivity of Circular Loop Antennas with Uniform Current
- Horn antennas Rectangular Horn Antennas

Suggested Reading

Text 3: "Antennas and Wave Propagation"

John D. Krauss, Ronald J Marhefka and Ahmad S Khan, 4th Special Indian Edition , McGraw- Hill Education Pvt. Ltd., 2010.

Text 7:

Sections: 7.1-7.8, 7.19, 7.20

Introduction

Definition:

A loop antenna is a radio antenna consisting of a loop or coil of wire, tubing, or other electrical conductor usually fed by a balanced source or feeding a balanced load

In general, a **loop antenna** is nothing but a radiating coil of any shape with one or more turns carrying a r.f. current. The loop antenna may assume any one of the shapes as shown in the Fig. 4.2.1. Generally a loop is formed on a ferrite core or air core.

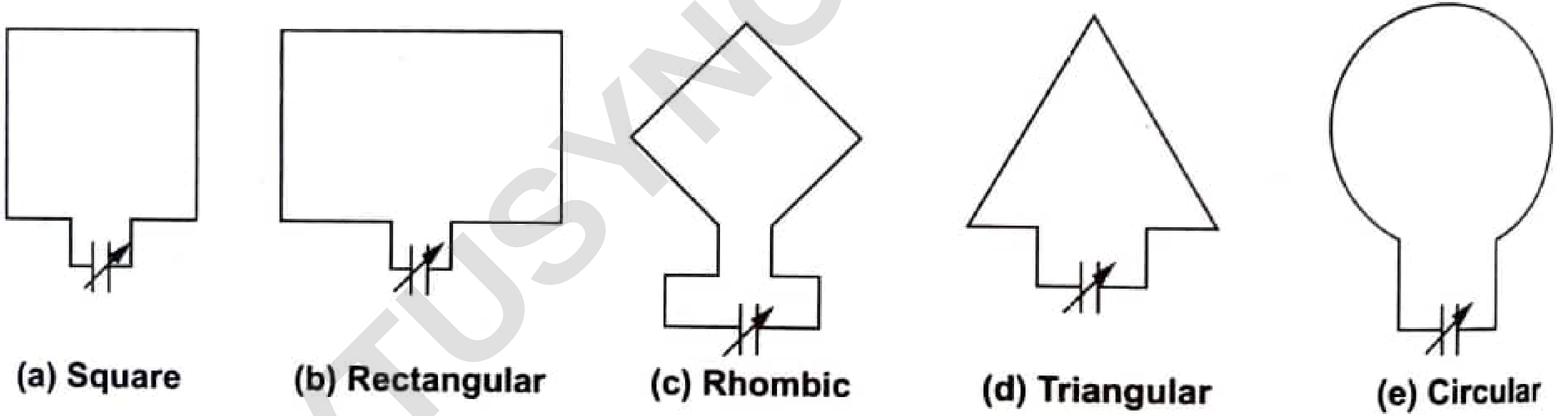


Fig. 4.2.1 Different shapes of loop antenna

Generally **a loop** may consists **one turn** or more turns on a **ferrite** or **air core**. If a loop consists more than one turn then it is commonly called as **frame.** Generally the loop antennas are of two types classified on the basis of the dimensions of the loop. In the first type, the dimensions of the loop are very small as compared with one wavelength. In other type, the dimensions of the loop are comparable with one wavelength. For such loop antennas the convenient assumption made is that the current in the loop has same magnitude and phase throughout.

Consider a rectangular loop with the turn with sides 23 and 14 as vertical arms and sides 12 and 34 as horizontal arms as shown in the Fig. 4.2.2.

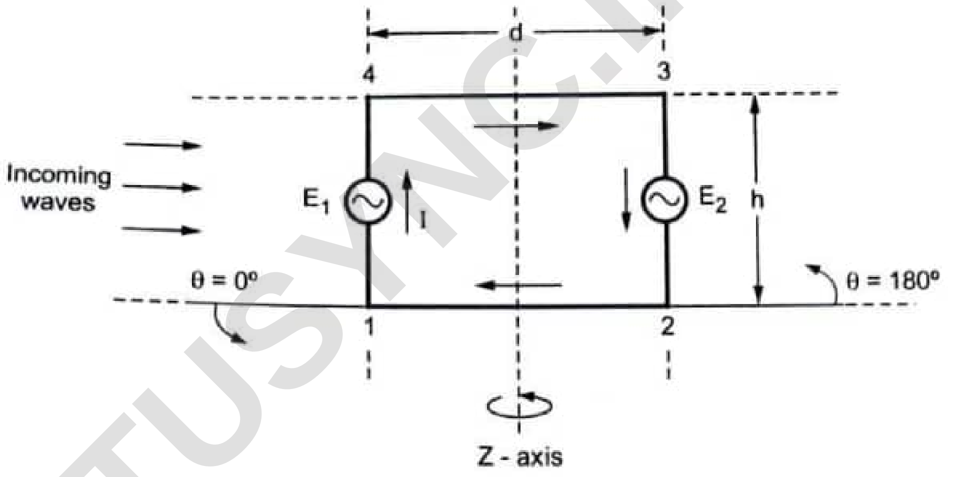


Fig. 4.2.2 Loop antenna

The horizontal arms and the vertical arms of the loop antenna acts as horizontal antenna and vertical antenna respectively.

The Small Loop

Field Pattern of Small Loop Antenna

For the transmission purposes, the radiation efficiency of a closed loop antenna is very low if the dimensions of the loop are very small as compared with the wavelength. The loop antenna discussed in the previous section is with small dimensions, so the magnitude and phase of the current throughout the loop remains same. So whenever a loop antenna is used for the transmission purposes, its dimensions are made comparable with the wavelength and the field pattern is analyzed by considering loop composed of four linear dipoles.

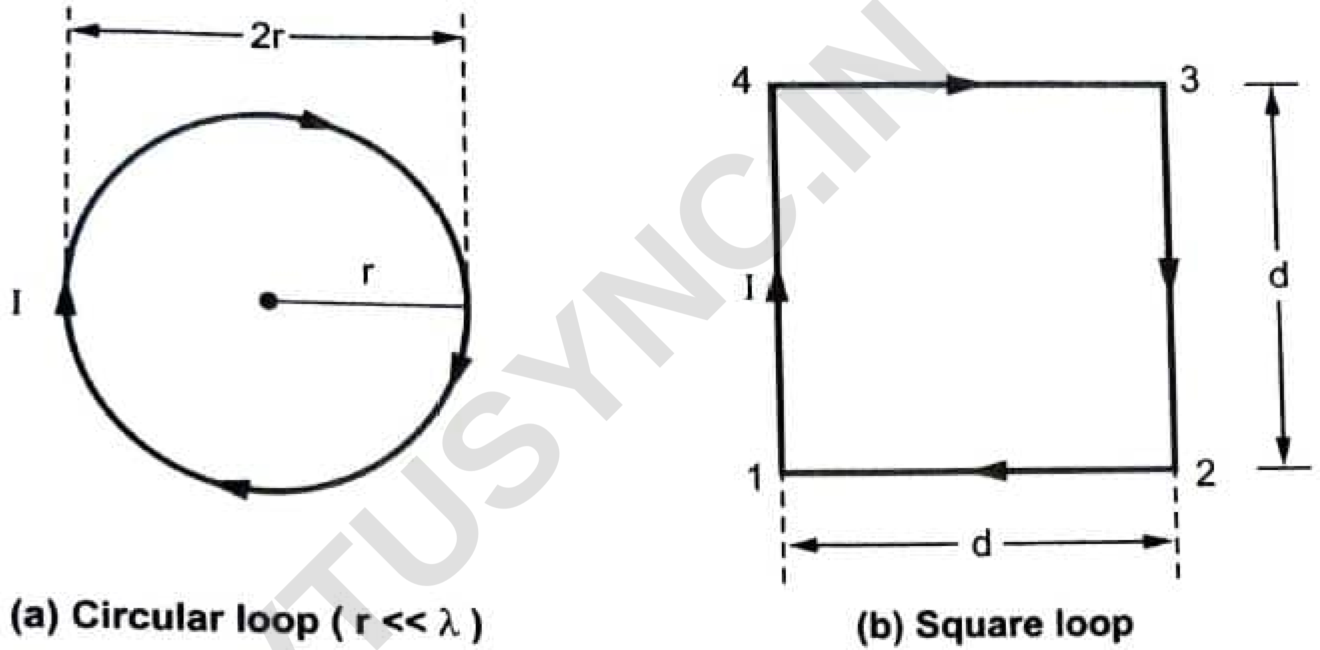
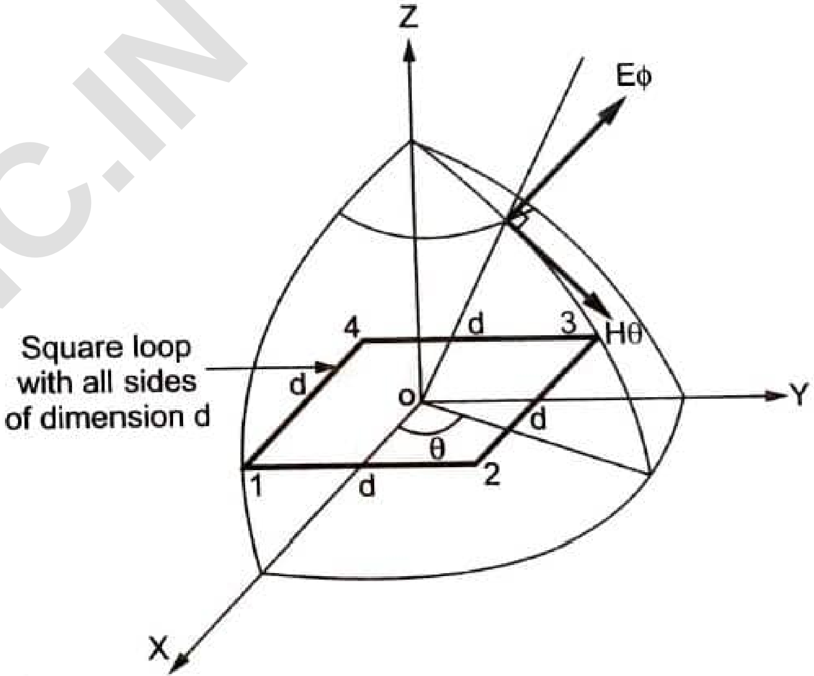


Fig. 4.3.1 Loop antenna in circular and square shape

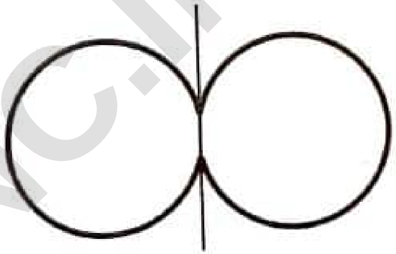
Consider a circular loop of radius r, very less compared with wavelength ($r << \lambda$) such that the current is in phase throughout the loop. It is represented in the Fig. 4.3.1 (a). Now consider that the circular loop is represented by a square loop of dimensions $d \times d$ such that the area of both loops is same. The square loop is illustrated in the Fig. 4.3.1 (b).

Consider that the square loop is located at the centre of the co-ordinate system as shown in the Fig. 4.3.2.

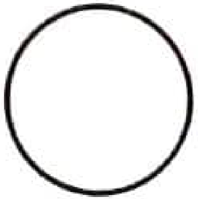


Then the far field of the square loop will have only E_ϕ component. To obtain the far field pattern, instead of considering all four short dipoles, it is sufficient to consider only two short dipoles such as sides 14 and 23.

Then the radiation pattern of these two short dipoles in the horizontal plane (i.e. X-Y plane) and the vertical plane (i.e. Y-Z plane) are as shown in the Fig. 4.3.3.



(a) Radiation pattern in horizontal plane



(b) Radiation pattern in vertical plane

Fig. 4.3.3 Radiation pattern of short dipole

From the Fig. 4.3.3 it is clear that the radiation pattern in the Y-Z plane for both the short dipoles is circular. In other words, we can say that the dipoles 1, 4 and 2, 3 are behaving like isotropic point sources in Y-Z plane as both are radiating uniformly in all the directions as shown in the Fig. 4.3.4.

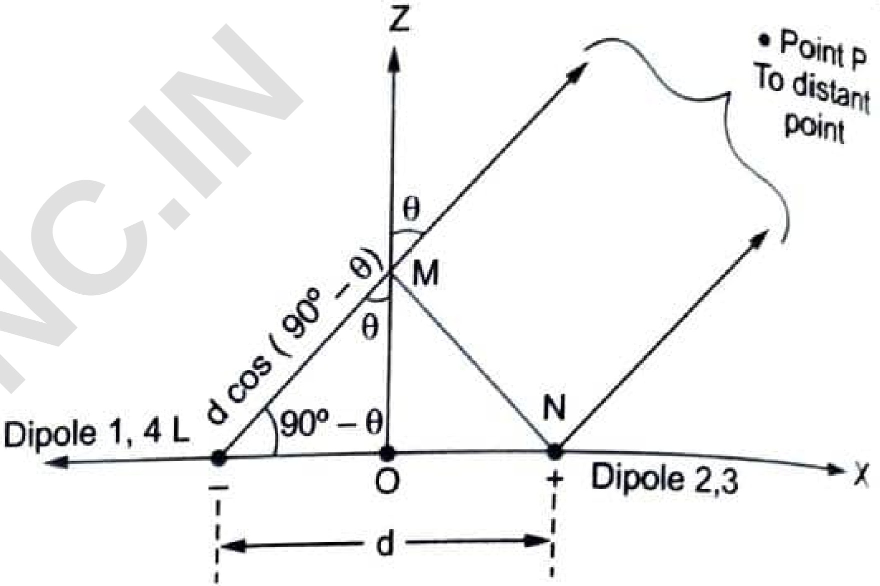


Fig. 4.3.4 Dipoles in Y-Z plane as isotropic point sources

Now the far field radiation due to two point sources with reference to centre O can be represented as,

$$E_{\phi} = \begin{matrix}\text{Field component} \\ \text{due to dipole 1, 4}\end{matrix} + \begin{matrix}\text{Field component} \\ \text{due to dipole 2, 3}\end{matrix}$$

From the Fig. 4.3.4 it is clear that the waves radiated from dipole 1, 4 will take more time to reach for point P as compared to the waves radiated from the dipole 2, 3. Because the waves radiated from the dipole 1, 4 will have to travel an extra distance from L to M as compared to that travelled by the waves radiated from the dipole 2, 3. This extra distance is called path difference.

Hence the path difference in above case is the distance from point L to M.

$$\therefore \quad \text{Path difference} = d\cos(90^\circ - \theta) \qquad \ldots (4.3.1)$$

The path difference interm of the wavelength can be expressed as,

$$\boxed{\text{Path difference} = \frac{d\cos(90^\circ - \theta)}{\lambda}} \qquad \ldots (4.3.2)$$

Let the phase angle be denoted by ψ. Then the phase angle and the path difference are related to each other through the expression given by,

$$\text{Phase angle} = \psi = 2\pi \times \text{Path difference}$$

$$\therefore \quad \boxed{\psi = \frac{2\pi}{\lambda} d\cos(90^\circ - \theta)} \qquad \ldots (4.3.3)$$

Both $\cos(90^\circ - \theta) = \sin\theta$, then the phase angle ψ is given by,

$$\boxed{\psi = \frac{2\pi}{\lambda} d\sin\theta} \qquad \ldots (4.3.4)$$

In general, the field component for any dipole is given by,

$$\boxed{\text{Field component} = \text{Magnitude} \times e^{j(\text{phase angle})}}$$

Let E_0 be the magnitude of electric field due to dipoles 1, 4 and 2, 3. Then the field component due to the dipole 1, 4 is given by,

$$E_{\phi 1} = -E_0 \cdot e^{j\frac{\psi}{2}} \qquad \dots (4.3.5)$$

Similarly the field component due to the dipole 2, 3 is given by,

$$E_{\phi 2} = +E_0 \cdot e^{-j\frac{\psi}{2}} \qquad \dots (4.3.6)$$

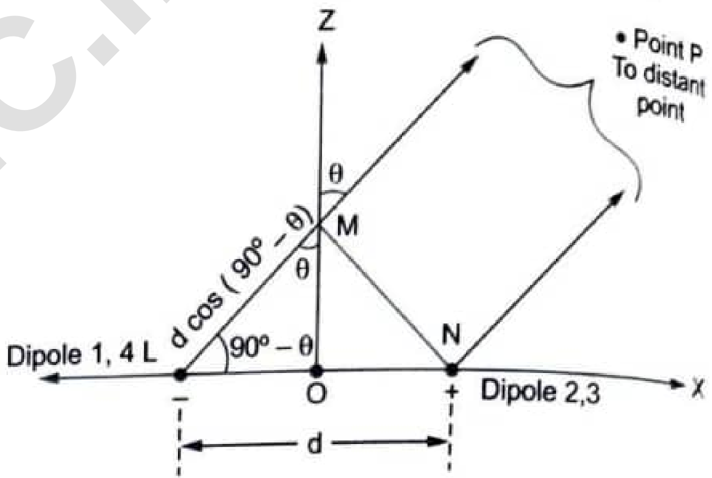
Hence far field radiation due to the square loop can obtained by adding equations (4.3.5) and (4.3.6) as,

$$E_\phi = E_{\phi_1} + E_{\phi_2}$$

$$\therefore \quad E_\phi = -E_0 \cdot e^{j\frac{\psi}{2}} + E_0 \cdot e^{-j\frac{\psi}{2}}$$

$$\therefore \quad E_\phi = -E_0 \left[e^{j\frac{\psi}{2}} - e^{-j\frac{\psi}{2}} \right]$$

$$\therefore \quad E_\phi = -j2E_0 \left[\frac{e^{j\frac{\psi}{2}} - e^{-j\frac{\psi}{2}}}{j2} \right]$$



But by the trigonometric identity,

$$\frac{e^{j\theta} - e^{-j\theta}}{j2} = \sin\theta$$

Hence we can write,

$$E_\phi = -j\,2\,E_0 \sin\frac{\psi}{2} \quad \ldots (4.3.7)$$

where ϕ = Phase angle = $\dfrac{2\pi}{\lambda} d \sin\theta$

But we know that, the phase shift β is given by,

$$\beta = \frac{2\pi}{\lambda} \quad \ldots (4.3.8)$$

Hence the phase angle expression can be modified as,

$$\psi = \frac{2\pi}{\lambda} d \sin\theta = \beta d \sin\theta \quad \ldots (4.3.9)$$

Substituting value of the phase angle in equation (4.3.7) we get,

$$\boxed{E_\phi = -j2\, E_0 \sin\left[\frac{\beta d \sin\theta}{2}\right] \text{ V/m}} \quad (4.3.10)$$

From equation (4.3.10), the other field component can be given by using following relationship. In the free space,

$$\frac{E_\phi}{H_\theta} = \eta_0 = 120\pi \qquad \dots (4.3.11)$$

Hence the magnetic field component is given by,

$$\therefore \quad H_\theta = \frac{E_\phi}{\eta_0}$$

$$\therefore \quad H_\theta = \frac{-j2}{120\pi} E_0 \cdot \sin\left[\frac{\beta d \sin\theta}{2}\right]$$

$$\therefore \quad \boxed{H_\theta = \frac{-j\,E_0}{60\pi} - \sin\left[\frac{\beta d \sin\theta}{2}\right] \text{ A/m}} \qquad \dots (4.3.12)$$

The magnitude of the individual field component is given by,

$$E_0 = \frac{j\,60\pi\,[I]\,L\sin 90^\circ}{r\lambda} = \frac{j\,60\pi\,[I]\,L}{r\lambda} \quad \ldots (4.3.13)$$

Here the angle $\theta = 90^\circ$ is considered because the field strength is measured from X-axis rather than from Z-axis.

Also in equation (4.3.10) if length of the side of square is very less than one wavelength ($d << \lambda$) then by trigonometric properties,

$$\sin\left[\frac{\beta d \sin\theta}{2}\right] \approx \frac{\beta d \sin\theta}{2} \quad \ldots (4.3.14)$$

VTUSYNC.IN

Hence using this concept, equation (4.3.10) can be written by putting value of E_0 from equation (4.3.13), we get,

$$E_\phi = -j2\left[\frac{j\,60\pi[I]L}{r\lambda}\right]\left(\frac{\beta d \sin\theta}{2}\right)$$

$$\therefore \quad E_\phi = \frac{60\pi\,[I]\,(d)}{r\lambda}\cdot\left(\frac{2\pi}{\lambda}\right)\cdot d\sin\theta \quad \ldots \left[\beta = \frac{2\pi}{\lambda} \text{ and } L = d \text{ for the square loop}\right]$$

$$\therefore \quad \boxed{E_\phi = \frac{120\pi^2\,[I]\,A\sin\theta}{r\lambda^2}} \quad \ldots (4.3.15)$$

where $A = d^2$ = Area of square loop of dimension $d \times d$

$[I]$ = Retarded current

Similarly the other component of the far field is given by,

$$\boxed{H_\theta = \frac{\pi\,[I]\sin\theta\,A}{r\lambda^2}} \quad \ldots (4.3.16)$$

Comparison of Far Fields of Small Loop and Short Dipole

Table Far fields of small electric dipoles and loops

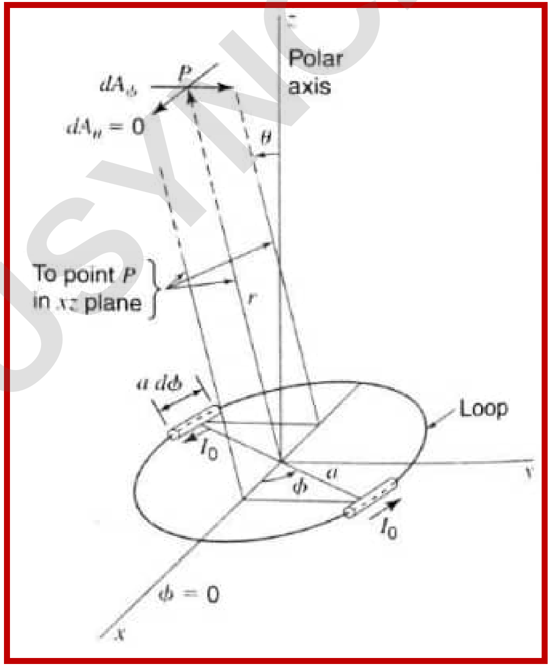
Field	Electric dipole	Loop
Electric	$E_\theta = \dfrac{j60\pi[I]\sin\theta}{r}\dfrac{L}{\lambda}$	$E_\phi = \dfrac{120\pi^2[I]\sin\theta}{r}\dfrac{A}{\lambda^2}$
Magnetic	$H_\phi = \dfrac{j[I]\sin\theta}{2r}\dfrac{L}{\lambda}$	$H_\theta = \dfrac{\pi[I]\sin\theta}{r}\dfrac{A}{\lambda^2}$

General Case of Loop Antenna

Consider a loop antenna with uniform, in-phase current and size not small; but comparable with one wavelength.

Consider a loop antenna with radius a located at the centre of co-ordinate system as shown in the Fig.

Assuming the current I uniform along the loop, the far field expressions can be obtained by finding the vector potential of the electric current.



The infinitesimal value of the ϕ - component of the vector potential A at point P (far away from the centre), due to the diametrically opposite infinitesimal dipoles is given by,

$$dA_{\phi} = \frac{\mu\, dM}{4\pi r} \quad \dots (4.5.1)$$

where dM = Current moment due to the pair of infinitesimal dipoles placed diametrically opposite with length $a d\phi$

In $\phi = 0$ plane, the ϕ - component of the retarted current moment due to one dipole is $[I](a d\phi) \cos\phi$.

where $[I] = I_0 \cdot e^{j\omega(t - \frac{r}{c})}$ and I_0 is the peak current.

Now consider the cross-section through the loop in x-z plane as shown in the Fig. 4.5.2.

From this Fig. 4.5.2 we can write the resultant moment at large distance r due to a pair of diametrically opposed dipole is given by,

$$dM = j2[I]\,(a d\phi)\cos\phi \sin\frac{\psi}{2} \qquad \ldots (4.5.2)$$

where ψ = Phase angle = $2\,\beta\, a \cos\phi \sin\theta$

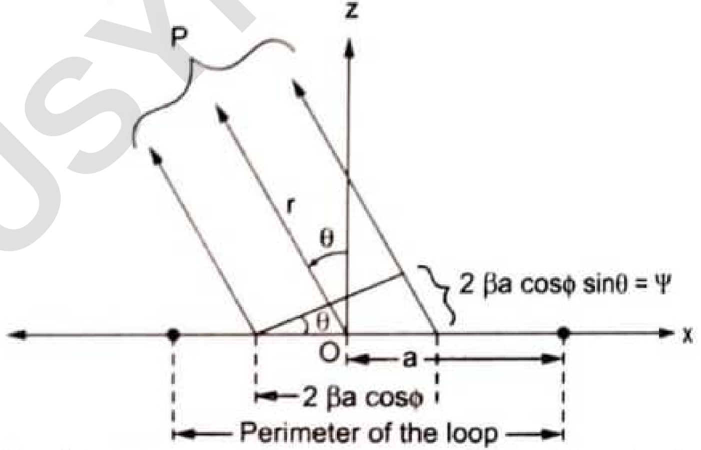


Fig. 4.5.2 Cross-section of loop with radius a in the x-z plane

Substituting value of phase angle ψ in equation (4.5.2), we get,

$$dM = j2\,[I]\,(a\,d\phi)\cos\phi\,\sin\left[\frac{2\beta a\cos\phi\sin\theta}{2}\right]$$

$$dM = j2\,[I]\,(a\,d\phi)\cos\phi\,\sin(\beta a\cos\phi\sin\theta) \quad \ldots (4.5.3)$$

Substituting value of dM in equation (4.5.1) and integrating both the sides with respect to corresponding variables,

$$\int dA_\phi = \frac{\mu}{4\pi R}\int_{\phi=0}^{\pi} j2[I]\,(a)\cos\phi\,\sin(\beta a\cos\phi\sin\theta)\,d\phi$$

$$A_\phi = \frac{\mu}{4\pi r}\{j2[I]\,a\}\int_0^{\pi}\sin(\beta a\cos\phi\sin\theta)\cos\phi\,d\phi$$

$$A_\phi = \frac{j\mu[I]\,a}{2\pi r}\int_0^{\pi}\sin(\beta a\cos\phi\sin\theta)\cos\phi\,d\phi \quad \ldots (4.5.4)$$

$$A_\phi = \frac{j\mu [I] a}{2\pi r} \int_0^\pi \sin(\beta a \cos\phi \sin\theta) \cos\phi \, d\phi \quad \ldots (4.5.4)$$

$$A_\phi = \frac{j\mu [I] a}{2r} J_1(\beta a \sin\theta) \quad \ldots (4.5.5)$$

Where $J_1(\beta a \sin\theta)$ is the **Bessel function** of the first order and of argument $(\beta a \sin\theta)$. The far electric field of the loop consists only the ϕ - component and it is given by,

$$E_\phi = -j\omega A_\phi \quad \ldots (4.5.6)$$

Substituting value of A_ϕ in equation (4.5.6), we get,

$$E_\phi = -j\omega \left\{ \frac{j\mu [I] a}{2r} J_1(\beta a \sin\theta) \right\}$$

$$E_\phi = \left\{ \frac{-j^2 \omega \mu [I] a}{2r} J_1(\beta a \sin\theta) \right\} \quad \ldots (4.5.7)$$

Put $j^2 = -1$ i.e. $-j^2 = +1$ and $\omega = 2\pi f$ in above equation,

We get,

$$E_\phi = \frac{2\pi f \mu [I] a}{2r} J_1(\beta a \sin\theta) \quad \dots (4.5.8)$$

Multiplying numerator and denominator terms by λ and combining λ with f in the numerator, we get,

$$E_\phi = \left(\frac{2\pi}{\lambda}\right)\frac{(f\lambda)\mu [I] a}{2r} J_1(\beta a \sin\theta) \quad \dots (4.5.9)$$

But by property $f \cdot \lambda = c$ i.e. velocity of light in the free space.

Substituting value of $f \cdot \lambda = c = 3 \times 10^8$ m/s and $\mu = 4\pi \times 10^{-7}$ H/m in equation (4.5.9), we get,

$$E_\phi = \left(\frac{2\pi}{\lambda}\right)\frac{(3\times10^8)(4\times\pi\times10^7)[I]\ a}{2r} J_1(\beta a \sin\theta)$$

$$\therefore \quad \boxed{E_\phi = \frac{60\pi(\beta)\,a\,[I]\,J_1(\beta a \sin\theta)}{r}} \qquad \dots (4.5.10)$$

The corresponding magnetic field component is given by,

$$H_\theta = \frac{E_\phi}{\eta_0} = \frac{60\pi(\beta)\,a\,[I]\,J_1(\beta a \sin\theta)}{r(120\pi)}$$

$$\therefore \quad \boxed{H_\theta = \frac{\beta[I]\ a\ J_1(\beta a \sin\theta)}{2r}} \qquad \dots (4.5.11)$$

Radiation Resistance of Loops

To find the radiation resistance of the loop antenna, the Poynting vector is integrated over large sphere and this total radiated power is equated to square of r.m.s. current and multiplied by radiation resistance R_{rad}.

$$\therefore \quad P = I_{rms}^2 \cdot R_{rad} = \frac{1}{2} I_M^2 R_{rad} \quad \dots (4.6.1)$$

The radiation resistance of the loop-antenna is the impedance appearing at the loop terminals where transmission line is connected as shown in the Fig. 4.6.1.

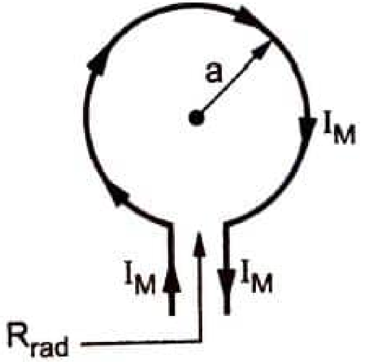


Fig. 4.6.1 R_{rad} of loop antenna

The average Poynting vector of a far field is given by

$$\bar{P} = \frac{1}{2} R_e(\bar{E} \times \bar{H}) \quad \dots (4.6.2)$$

Hence radial component due to E_θ and H_ϕ is given by,

$$P_r = \frac{1}{2} R_e(E_\theta H_\phi^*) \text{ where } H_\phi^* \text{ conjugate of } H_\phi$$

But $$E_\theta = \eta_0 H_\phi, \text{ we can write}$$

$$\therefore \quad P_r = \frac{1}{2} R_e(\eta_0 H_\phi H_\phi^*)$$

$$\therefore \quad \boxed{P_r = \frac{1}{2} |H_\phi|^2 \eta_0} \qquad \ldots (4.6.3)$$

But according to the discussion in the previous section, from equation (4.5.11),

$$H_\theta = \frac{B_a [I]}{2r} J_1(\beta a \sin\theta)$$

Using this value in equation (4.6.3), we get,

$$P_r = \frac{1}{2}\left|\frac{\beta a[I]}{2r} J_1(\beta a \sin\theta)\right|^2 \cdot (120\pi) \qquad \ldots \eta_0 = 120\pi\Omega$$

$$\therefore \quad P_r = \frac{1}{2}(120\pi)\frac{(\beta a I_M)^2}{4r^2} J_1^2(\beta a \sin\theta)$$

$$\therefore \quad \boxed{P_r = \frac{15\pi(\beta a I_M)^2}{r^2} J_1^2(\beta a \sin\theta)} \qquad \ldots (4.6.4)$$

To find total power radiated integrating this power over a large sphere, we can write,

$$P = \iint \bar{P} \cdot d\bar{s}$$

$$= \int_0^{2\pi}\int_0^{\pi} \frac{15\pi(\beta a I_M)^2}{r^2} J_1^2(\beta a \sin\theta) \cdot r^2 \sin\theta \, d\theta \, d\phi$$

To find total power radiated integrating this power over a large sphere, we can write,

$$P = \iint \bar{P} \cdot d\bar{s}$$

$$= \int_0^{2\pi} \int_0^{\pi} \frac{15\pi (\beta a I_M)^2}{r^2} J_1^2 (\beta a \sin\theta) \cdot r^2 \sin\theta \, d\theta \, d\phi$$

$$\therefore \quad P = [\phi]_0^{2\pi} \left\{ 15\pi (\beta a I_M)^2 \int_0^{\pi} J_1^2 (\beta a \sin\theta) \sin\theta \, d\theta \right\}$$

$$\therefore \quad P = (2\pi)(15\pi)(\beta a I_M)^2 \int_0^{\pi} J_1^2 (\beta a \sin\theta) \sin\theta \, d\theta$$

$$\therefore \quad \boxed{P = 30\pi^2 (\beta a I_M)^2 \int_0^{\pi} J_1^2 (\beta a \sin\theta) \sin\theta \, d\theta} \qquad \ldots (4.6.5)$$

But for a loop which is smaller interms of λ, we can write,

$$J_1^2(\beta a \sin\theta) = \left(\frac{\beta a \sin\theta}{2}\right)^2$$

Using this property, equation (4.6.5) can be modified as,

$$P = 30\pi^2(\beta a I_M)^2 \int_0^{\pi} \left(\frac{\beta a \sin\theta}{2}\right)^2 \cdot \sin\theta \, d\theta$$

$$\therefore \quad P = \frac{30\pi^2}{4}(\beta a I_M)^2(\beta a)^2 \int_0^{\pi} \sin^2\theta \sin\theta \, d\theta$$

Solving integral term and putting limits we get,

$$\therefore \quad \boxed{P = 10\pi^2(\beta a)^4 I_M^2} \qquad \ldots (4.6.6)$$

Assuming ideal condition i.e. zero loss, we can equate this equation with equation (4.6.1), we get,

$$\frac{1}{2} I_M^2 R_{rad} = 10\pi^2 (\beta a)^4 I_M^2$$

$$\therefore \quad R_{rad} = 20\pi^2 (\beta a)^4$$

$$\therefore \quad R_{rad} = 20\pi^2 \beta^4 a^4$$

$$\therefore \quad R_{rad} = 20\beta^4 (\pi a^2)^2$$

But πa^2 is the area A of the loop. Hence radiation, resistance is given by,

$$\therefore \quad \boxed{R_{rad} = 20\beta^4 A^2 \ \Omega} \qquad \dots (4.6.7)$$

But $\beta = \frac{2\pi}{\lambda}$, hence equation (4.6.7) becomes,

$$R_{rad} = 20\left(\frac{2\pi}{\lambda}\right)^4 A^2 = 31171\left(\frac{A^2}{\lambda^4}\right)\Omega$$

Hence the approximate formula is given by,

$$\therefore \quad \boxed{R_{rad} = 31200\left(\frac{A}{\lambda^2}\right)^2 \Omega} \qquad \ldots (4.6.8)$$

The same expression can be expressed in the following form,

$$\therefore \quad \boxed{R_{rad} = 320\pi^4\left(\frac{A}{\lambda^2}\right)^2 \Omega} \qquad \ldots (4.6.9)$$

All the above expressions are for the loop with a single turn. If the loop consists N turns, then the radiation resistance is N^2 times the radiation resistance for one turn loop.

$$\therefore \quad \boxed{R_{rad} = 31200\left(\frac{A}{\lambda^2}\right)^2 N^2 = 320\,\pi^4\left(\frac{A}{\lambda^2}\right)^2 N^2\,\Omega} \qquad \ldots (4.6.10)$$

Let C be the circumference of a loop with radius a. Then we can write, $C = 2\pi a$. Equation (4.6.7) can be rearranged by putting value of β in terms of λ as,

$$R_{rad} = 20\left(\frac{2\pi}{\lambda}\right)^2 A^2$$

$$\therefore \quad R_{rad} = 20\left(\frac{2\pi}{\lambda}\right)^2 (\pi a^2)^2$$

$$\therefore \quad R_{rad} = 20\pi^2\left(\frac{2\pi a}{\lambda}\right)^4$$

$$\therefore \quad \boxed{R_{rad} = 20\pi^2\left(\frac{C}{\lambda}\right)^4 \Omega} \qquad \ldots (4.6.11)$$

Putting value of π^2, we get approximate expression for R_{rad} as,

$$\therefore \quad \boxed{R_{rad} \approx 197\left(\frac{C}{\lambda}\right)^4 \Omega} \qquad \ldots (4.6.12)$$

All above expressions of the radiation resistance are for the loop with radius less than λ. Now in general to find radiation resistance of a circular loop with any radius a, we write,

$$P = 30\pi^2 (\beta a I_M)^2 \int_0^{\pi} J_1^2 (\beta a \sin\theta) \sin\theta \, d\theta$$

Now by property of Bessel functions, we can write,

$$\therefore \quad \boxed{\int_0^{\pi} J_1^2 (x \sin\theta) \sin\theta \, d\theta = \frac{1}{x} \int_0^{2x} J_2(y) dy}$$

Hence the expression for the power can be written as,

$$P = 30\pi^2 (\beta a I_M)^2 \cdot \frac{1}{\beta a} \int_0^{2\beta a} J_2(y) \, dy \qquad \dots (4.6.13)$$

When $\frac{C}{\lambda} \geq 5$, the loop is considered as a large loop. Let us use an approximation given by,

$$\int_0^{\pi} J_1^2(\beta a \sin\theta)\sin\theta\, d\theta = \frac{1}{\beta a}\int_0^{2\beta a} J_2(y)dy \approx \frac{1}{\beta a}\cdot 1 \approx \frac{1}{\beta a}$$

Then equation (4.6.13) becomes,

$$P = 30\pi^2(\beta a I_M)^2 \cdot \frac{1}{\beta a}$$

$$\therefore \quad P = 30\pi^2(\beta a) I_M^2 \qquad \ldots (4.6.14)$$

But $\beta = \frac{2\pi}{\lambda}$

$$\therefore \quad P = 30\pi^2\left(\frac{2\pi}{\lambda}a\right)I_M^2$$

$$\ldots \left(\because \frac{2\pi}{\lambda}a = \frac{2\pi a}{\lambda} = \frac{\pi D}{\lambda} = \frac{C}{\lambda}\right)$$

D = diameter = 2a

$$\therefore \quad P = 30\pi^2 I_M^2\left(\frac{C}{\lambda}\right) \qquad \ldots (4.6.15)$$

Similar to previous condition, equating equation (4.6.15) with equation (4.6.1), we get

$$\frac{1}{2} I_M^2 R_{rad} = 30\pi^2 I_M^2 \left(\frac{C}{\lambda}\right)$$

$$\therefore \quad \boxed{R_{rad} = 60\pi^2 \left(\frac{C}{\lambda}\right) \Omega} \qquad \ldots (4.6.16)$$

Putting value of $\pi = 3.142$, we get,

$$R_{rad} = 60(3.142)^2 \left(\frac{C}{\lambda}\right)$$

$$\therefore \quad \boxed{R_{rad} \simeq 592 \left(\frac{C}{\lambda}\right) \Omega} \qquad \ldots (4.6.17)$$

Let us use C = $2\pi a$ in above expression, then we get,

$$R_{rad} = 592\left(\frac{2\pi a}{\lambda}\right)$$

$$\therefore \quad R_{rad} = 592\times 2\times 3.142\left(\frac{a}{\lambda}\right)$$

$$\therefore \quad \boxed{R_{rad} \approx 3720\left(\frac{a}{\lambda}\right)\Omega} \qquad \dots (4.6.18)$$

Directivity of Circular Loop Antenna

In general, the directivity of an antenna is defined as the ratio of maximum radiation intensity to the average radiation intensity.

$$\therefore \quad \boxed{D = \frac{\text{Maximum radiation intensity}}{\text{Average radiation intensity}} = \frac{(P_r)(r^2)}{\left(\frac{P}{4\pi}\right)}} \qquad \dots (4.7.1)$$

Using equations (4.6.4) and (4.6.5) from section 4.6 for P_r and P respectively, the directivity can be written as,

$$D = \frac{\left\{\frac{15\pi(\beta a I_m)^2}{r^2} J_1^2(\beta a \sin\theta)\right\}_{max} r^2}{\left\{\frac{30\pi^2 (\beta a I_m)^2 \int_0^{\pi} J_1^2(\beta a \sin\theta)\sin\theta\, d\theta}{4\pi}\right\}}$$

$$\therefore \quad D = \frac{[60\pi^2 (\beta a I_m)^2 J_1^2 (\beta a \sin\theta)]_{max}}{30\pi^2 (\beta a I_m)^2 \int_0^{\pi} J_1^2 (\beta a \sin\theta) \sin\theta \, d\theta}$$

$$\therefore \quad D = \frac{2 [J_1^2 (\beta a \sin\theta)]_{max}}{\frac{1}{\beta a} \int_0^{2\beta a} J_1^2 (\beta a \sin\theta) \sin\theta \, d\theta}$$

$$\therefore \quad D = \frac{2\beta a [J_1^2 (\beta a \sin\theta)]_{max}}{\int_0^{2\beta a} J_1^2 (\beta a \sin\theta) \sin\theta \, d\theta} \quad \ldots (4.7.2)$$

But $\beta a = \frac{C}{\lambda}$

$$\therefore \qquad D = \frac{2\frac{C}{\lambda}\left[J_1^2\left(\frac{C}{\lambda}\sin\theta\right)\right]_{max}}{\int_0^{\frac{2C}{\lambda}} J_1^2\left(\frac{C}{\lambda}\sin\theta\right)\sin\theta\, d\theta}$$

$$\therefore \qquad D = \frac{2\frac{C}{\lambda}\left[J_1^2\left(\frac{C}{\lambda}\sin\theta\right)\right]_{max}}{\int_0^{2\frac{C}{\lambda}} J_2(y)\, dy} \qquad \text{... (4.7.3)}$$

This expression is called Foster's expression for directivity of a circular loop of any circumference $\left(\frac{C}{\lambda}\right)$ and uniform in-phase current. It is important that the angle θ is the angle for which the field is maximum.

Case I : Small loop $\frac{C}{\lambda} < \frac{1}{3}$

For condition $\frac{C}{\lambda} < \frac{1}{3}$, the directivity of loop is given by,

$$D = \frac{2\,[J_1^2\,(\beta\, a \sin\theta)]_{max}}{\int_0^{\pi} J_1^2\,(\beta\, a \sin\theta)\sin\theta\, d\theta} \quad \text{... (4.7.4)}$$

For small loop, $J_1(\theta) = \frac{\theta}{2}$ i.e. $J_1^2(\theta) = \left[\frac{\theta}{2}\right]^2$

$$\therefore \quad D = \frac{2\left[\frac{\beta\, a \sin\theta}{2}\right]^2_{max}}{\int_0^{\pi}\left[\frac{\beta\, a \sin\theta}{2}\right]^2 \sin\theta\, d\theta}$$

$$\therefore \quad D = \frac{2\left[\left(\frac{\beta a}{2}\right)^2 \sin^2\theta\right]_{max}}{\left(\frac{\beta a}{2}\right)^2 \int_0^{\pi} \sin^2\theta \cdot \sin\theta \, d\theta}$$

$$\therefore \quad D = \frac{2\,[\sin^2\theta]_{max}}{\int_0^{\pi} \sin^3\theta \, d\theta}$$

$$\therefore \quad D = \frac{2\,[\sin^2\theta]_{max}}{2\int_0^{\pi/2} \sin^3\theta \, d\theta} = \frac{[\sin^2\theta]_{max}}{\int_0^{\pi/2} \sin^3\theta \, d\theta} \qquad \ldots (4.7.5)$$

When $\theta = 90^\circ$, field is maximum and $[\sin^2\theta] = [\sin 90^\circ]^2_{max} = 1$ and $\int_0^{\pi/2} \sin^3\theta \, d\theta = \frac{2}{3}$

$$\therefore \quad \boxed{D = \frac{3}{2}} \qquad \ldots (4.7.6)$$

Note that the pattern of short dipole is same as for the small loop and thus the value of directivity is also same for the loop as well as short dipole.

Case II : Large loop $\frac{C}{\lambda} \geq 5$

For a large loop, the directivity is given by,

$$D = \frac{2\left(\frac{C}{\lambda}\right)\left[J_1^2\left(\frac{C}{\lambda}\sin\theta\right)\right]_{max}}{\int_0^{\frac{2C}{\lambda}} J_2(y)\,dy} \quad \ldots (4.7.7)$$

But $\int_0^{\frac{2C}{\lambda}} J_2(y)\,dy \approx 1$

$$\therefore \quad D \approx 2\frac{C}{\lambda}\left[J_1^2\left(\frac{C}{\lambda}\sin\theta\right)\right]_{max}$$

But according to condition studied for $\frac{C}{\lambda} \geq 1.84$, the maximum value of $J_1\left(\frac{C}{\lambda}\sin\theta\right)$ equals 0.584.

Then, $$D = 2\frac{C}{\lambda}[0.584]^2 = 0.68211\left(\frac{C}{\lambda}\right)$$

$$\therefore \quad \boxed{D \approx 0.682\left(\frac{C}{\lambda}\right)} \quad \ldots (4.7.8)$$

Example 4.7.1 *Find radiation resistance of a circular loop antenna of radius 0.3183 m operating at 1 MHz. The radius of a wire used is 0.4 mm, conductivity of the wire is 57 mS/m and* $\mu_r = 1.0$.

Solution : For a circular loop, radiation resistance is given by,

$$R_{rad} = 197\left(\frac{C}{\lambda}\right)^4$$

Example 4.7.1 *Find radiation resistance of a circular loop antenna of radius 0.3183 m operating at 1 MHz. The radius of a wire used is 0.4 mm, conductivity of the wire is 57 mS/m and* $\mu_r = 1.0$.

Solution : For a circular loop, radiation resistance is given by,

$$R_{rad} = 197\left(\frac{C}{\lambda}\right)^4$$

where

$$C = 2\pi a = 2\times\pi\times 0.3183 = 2\text{ m}$$

$$\lambda = \frac{c}{f} = \frac{3\times10^8}{1\times10^6} = 300\text{ m}$$

Hence

$$R_{rad} = 197\left(\frac{2}{300}\right)^4 = 0.3891\times 10^{-6}\,\Omega = 0.3891\,\mu\Omega$$

Example 4.7.2 *The diameter of a circular loop antenna is 0.04 λ. How many turns of the antenna will give a radiation resistance of 36 Ω ?*

$$\text{Area of loop} = A = \pi a^2 = \pi\left(\frac{d}{2}\right)^2 = \pi\left(\frac{0.04\lambda}{2}\right)^2 = 0.001256\lambda^2$$

But if $A < \frac{\lambda^2}{100}$ i.e. $\frac{A}{\lambda^2} = 0.01$, the loop is supposed to be a small loop.

Example 4.7.2 *The diameter of a circular loop antenna is 0.04 λ. How many turns of the antenna will give a radiation resistance of 36 Ω ?*

Solution :

$$\text{Area of loop} = A = \pi a^2 = \pi\left(\frac{d}{2}\right)^2 = \pi\left(\frac{0.04\lambda}{2}\right)^2 = 0.001256\lambda^2$$

But if $A < \frac{\lambda^2}{100}$ i.e. $\frac{A}{\lambda^2} = 0.01$, the loop is supposed to be a small loop.

For given loop, $\frac{A}{\lambda^2} = 0.001256 < 0.01$, thus loop is considered to be small loop.

For small loop, radiation resistance is given by,

$$R_{rad} = 31200\left(\frac{A}{\lambda^2}\right)^2 N^2$$

$$\therefore \quad 36 = 31200\left[\frac{0.001256\lambda^2}{\lambda^2}\right]^2 N^2$$

$$\therefore \quad (0.001256)^2 N^2 = \frac{36}{31200}$$

$$\therefore \quad (0.001256)^2 N^2 = 0.001154$$

$$\therefore \quad N^2 = 731.5205$$

$$\therefore \quad N = 27.04 \approx 27$$

Example 4.7.4 *Find radiation resistance of a loop antenna with diameter 0.5 m operating at 1 MHz.*

In general, radiation resistance of a loop antenna with any radius a is given by,

$$R_{rad} = 3720 \left(\frac{a}{\lambda} \right)$$

Example 4.7.4 *Find radiation resistance of a loop antenna with diameter 0.5 m operating at 1 MHz.*

Solution : **Given :** Diameter = d = 0.5 m, Radius = a = $\frac{d}{2}$ = 0.25 m

$$f = 1 \text{ MHz} = 1 \times 10^6 \text{ Hz}$$

Hence
$$\lambda = \frac{c}{f} = \frac{3\times10^8}{1\times10^6} = 300 \text{ m}$$

In general, radiation resistance of a loop antenna with any radius a is given by,

$$R_{rad} = 3720\left(\frac{a}{\lambda}\right) = 3720 \times \left(\frac{0.25}{200}\right) = 3.1\ \Omega$$

Example 4.7.5 *Determine directivity of loop antenna having radius of 1.0 m when it is operated at 0.9 MHz.*

Solution :

$$a = 1.0 \text{ m}$$

$$f = 0.9 \text{ MHz} = 0.9 \times 10^6 \text{ Hz}$$

$$\therefore \quad \lambda = \frac{c}{f} = \frac{3\times10^8}{0.9\times10^6} = 333.3333 \text{ m}$$

Let us calculate value of $\frac{C}{\lambda}$

$$\frac{C}{\lambda} = \frac{2\pi a}{\lambda} = \frac{2\times\pi\times1}{333.3333} = 0.01885 < \frac{1}{3}$$

Hence as $\frac{C}{\lambda} < \frac{1}{3}$, the directivity of loop antenna is directly given by,

$$D = \frac{3}{2} = 1.5$$

Example 4.7.6 *Find directivity and radiation resistance of a loop antenna with diameter of 1.5 λ.*

Solution : $d = \text{Diameter} = 1.5\,\lambda$

$$a = \text{Radius} = \frac{d}{2} = \frac{1.5\lambda}{2} = 0.75\,\lambda$$

Now
$$\frac{C}{\lambda} = \frac{2\pi a}{\lambda} = \frac{\pi d}{\lambda} = \frac{\pi(1.5\lambda)}{\lambda} = 1.5\,\pi > \frac{1}{3}$$

Hence for such large loop antenna, radiation resistance is given by,

$$R_{rad} = 3720\left(\frac{a}{\lambda}\right) = 3720\left(\frac{0.75\lambda}{\lambda}\right) = \mathbf{2790\ \Omega}$$

Similarly the directivity of such large loop antenna is given by,

$$D = 0.682\left(\frac{C}{\lambda}\right) = 0.682\,(1.5\,\pi) = \mathbf{3.21385}$$

Example 4.7.8 *The radius of a circular loop antenna is 0.02 λ. How many turns of the antenna will give a radiation resistance of 35 Ω ?*

Solution : Given loop antenna has circular loop with radius 0.02 λ. The loop is said to be small loop if,

$$A < \frac{\lambda^2}{100}$$

For given loop, $A = \pi r^2 = \pi (0.02\lambda)^2 = 1.2566 \times 10^{-3} \lambda^2$

$$\therefore \quad \frac{A}{\lambda^2} = 0.001256 \text{ which is less than } 0.01.$$

Hence the given loop is small loop.

Now for a small loop, with N turns, the radiation resistance is given by,

$$R_{rad} = 31200\left(\frac{AN}{\lambda^2}\right)^2$$

$$\therefore \quad 35 = 31200\left(\frac{1.2566\times 10^{-3}\,\lambda^2 \times N^2}{\lambda^2}\right)^2$$

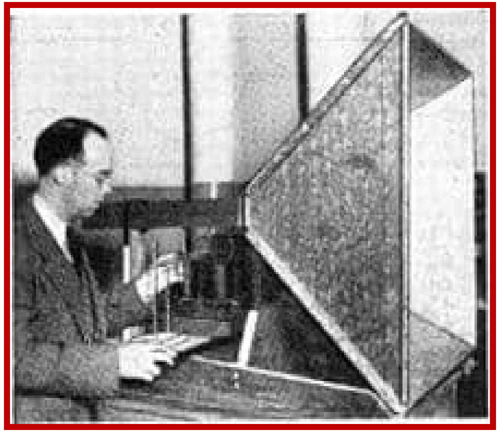
$$\therefore (1.2566\times10^{-3} N^2) = \sqrt{\frac{35}{31200}}$$

$$\therefore \quad N = 5.1627$$

Thus $\quad N = 5$

Thus for a 0.02 λ radius circular loop, 5 turns of loop will give radiation resistance of 35 Ω.

Horn antennas



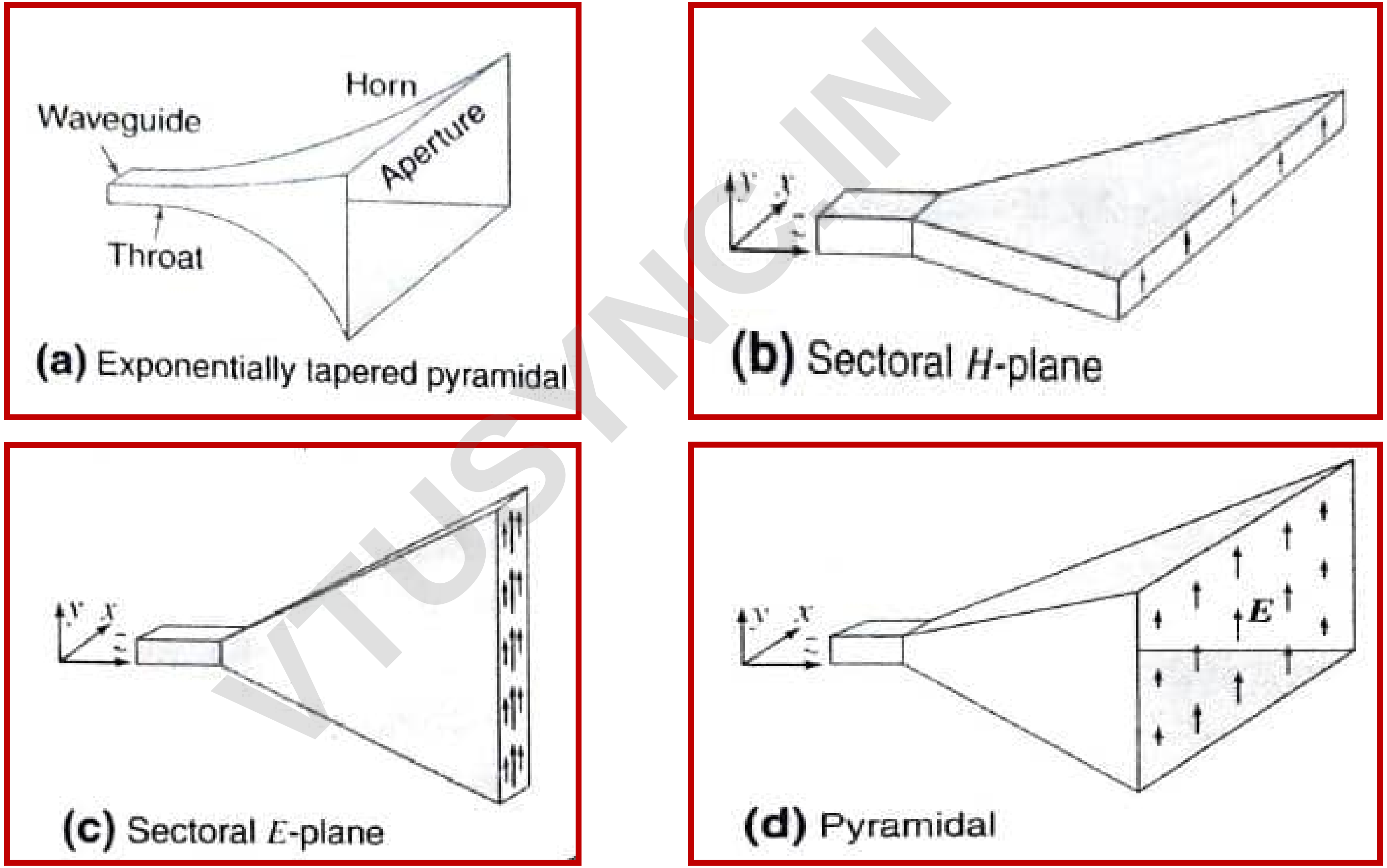
The first modern horn antenna was invented by Wilmer L. Barrow in 1938

A horn antenna may be regarded as a flared-out (or opened-out) waveguide. The function of the horn is to produce a uniform phase front with a larger aperture than that of the waveguide and hence greater directivity. Horn antennas are not new. Jagadis Chandra Bose constructed a pyramidal horn in 1897.

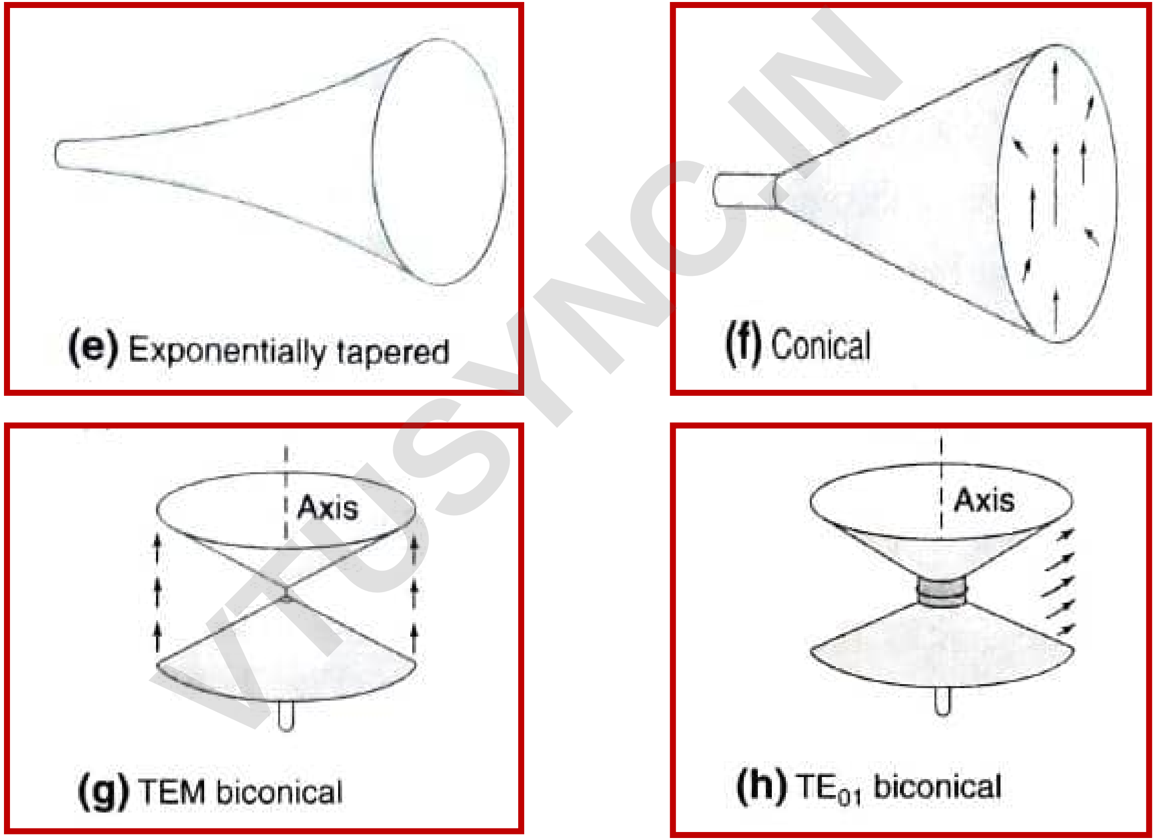
A horn antenna or microwave horn is an antenna that consists of a flaring metal waveguide shaped like a horn to direct radio waves in a beam. Horns are widely used as antennas at UHF and microwave frequencies, above 300 MHz

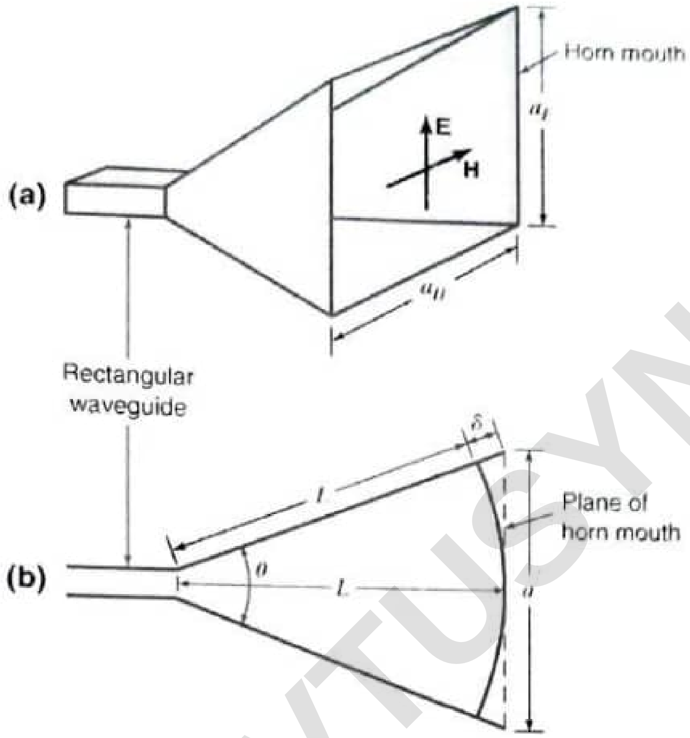
A horn antenna is used to transmit radio waves from a waveguide (a metal pipe used to carry radio waves) out into space, or collect radio waves into a waveguide for reception.

Rectangular Horns



Circular Horns





From Fig.

$$\cos\frac{\theta}{2} = \frac{L}{L+\delta} \qquad (1)$$

$$\sin\frac{\theta}{2} = \frac{a}{2(L+\delta)} \qquad (2)$$

$$\tan\frac{\theta}{2} = \frac{a}{2L} \qquad (3)$$

where

θ = flare angle (θ_E for E plane, θ_H for H plane), deg

a = aperture (a_E for E plane, a_H for H plane), m

L = horn length, m

δ = path length difference, m

Figure (*a*) Pyramidal horn antenna. (*b*) Cross section with dimensions used in analysis. The diagram can be for either E-plane or H-plane cross sections. For the E plane the flare angle is θ_E and aperture a_E. For the H plane the flare angle is θ_H and the aperture a_H.

It follows that the maximum directivity occurs at the largest flare angle for which δ does not exceed a certain value (δ_0). Thus, from (1) the *optimum horn dimensions* can be related by

or

Optimum horn dimensions	$$\delta_0 = \frac{L}{\cos(\theta/2)} - L = \text{optimum } \delta$$	(6)
	$$L = \frac{\delta_0 \cos(\theta/2)}{1 - \cos(\theta/2)} = \text{optimum length}$$	(7)

The Rectangular Horn Antenna

The directivity of a horn antenna

The directivity (or gain, assuming no loss) of a horn antenna can be expressed in terms of its effective aperture. Thus,

$$D = \frac{4\pi A_e}{\lambda^2} = \frac{4\pi \varepsilon_{ap} A_p}{\lambda^2} \qquad (1)$$

where

A_e = effective aperture, m^2
A_p = physical aperture, m^2
ε_{ap} = aperture efficiency = A_e/A_p
λ = wavelength, m

For a rectangular horn $A_p = a_E a_H$ and for a conical horn $A_p = \pi r^2$, where r = aperture radius. It is assumed that a_E, a_H or r are all at least 1λ. Taking $\varepsilon_{ap} \simeq 0.6$, (1) becomes

$$D \simeq \frac{7.5 A_p}{\lambda^2} \tag{2}$$

or

$$D \simeq 10 \log\left(\frac{7.5 A_p}{\lambda^2}\right) \qquad \text{(dBi)} \tag{3}$$

For a pyramidal (rectangular) horn (3) can also be expressed as

$$D \simeq 10 \log(7.5 a_{E\lambda} a_{H\lambda}) \tag{4}$$

where

$a_{E\lambda}$ = E-plane aperture in λ
$a_{H\lambda}$ = H-plane aperture in λ

CBCS SCHEME

USN

15EC71

Seventh Semester B.E. Degree Examination, Dec.2018/Jan.2019

Microwaves and Antennas

Time: 3 hrs. Max. Marks: 80

Note: Answer any FIVE full questions, choosing ONE full question from each module.

Module-5

9 a. Derive the expression for strength $E\phi$ and $H\theta$ in case of small loop. (06 Marks)

b. Explain the working and design considerations of Log-periodic antenna. (06 Marks)

c. A 16-turn helical beam antenna has a circumference of λ and turn spacing of $\lambda/4$. Find : i) HPBW ii) axial ratio iii) directivity. (04 Marks)

OR

10 a. Show that the radiation resistance of small loop is $31171\left(\frac{A}{\lambda^2}\right)^2$. (05 Marks)

b. Write a short notes on : i) Yagi Uda array ii) parabolic reflector. (06 Marks)

c. Determine the length L, H-plane aperture and flare angles θ_E and θ_H of a pyramidal horn for which the E-plane aperture $a_E = 10\lambda$. Let $\delta = 0.2\lambda$ in the E-plane and 0.375λ in the H-plane. Also determine beam widths and directivity. (05 Marks)

CBCS SCHEME

USN | | | | | | | | | | 15EC71

Seventh Semester B.E. Degree Examination, June/July 2019

Microwave and Antennas

Time: 3 hrs. Max. Marks: 80

Module-5

9 a. Obtain the expression for radiation resistance of small loop antenna. **(08 Marks)**

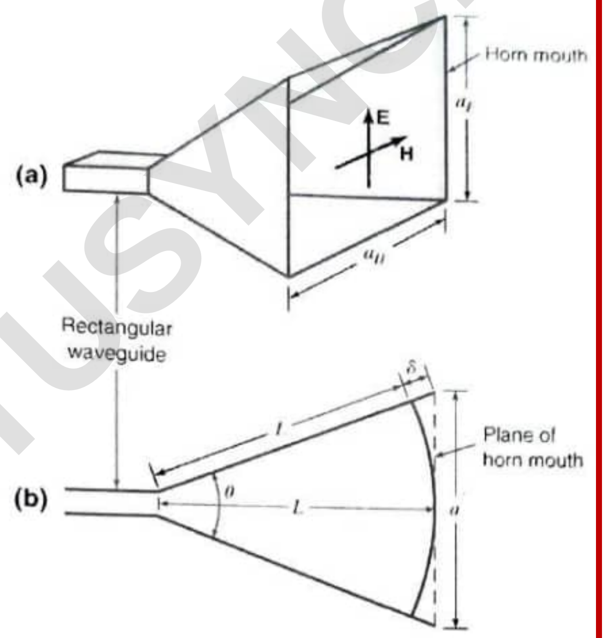
b. With neat diagram explain the operation of log-periodic antenna. **(08 Marks)**

OR

10 a. Determine the length L_1 H-plane aperture and flare angle θ_E and θ_H of a pyramidal horn for which the E-plane aperture $a_E = 10\lambda$. The horn is fed by a rectangular waveguide with TE_{10} mode. Let $\delta = 0.2\lambda$ in the E plane and 0.375λ in the H plane. Also find what are beam widths and what is the directivity. **(08 Marks)**

b. Discuss the following antenna types (i) Helical Antenna (ii) Yagi-uda-array. **(08 Marks)**

EXAMPLE (a) Determine the length L, H-plane aperture and flare angles θ_E and θ_H (in the E and H planes, respectively) of a pyramidal horn as in Fig. 7–40d for which the E-plane aperture $a_E = 10\lambda$. The horn is fed by a rectangular waveguide with TE_{10} mode. Let $\delta = 0.2\lambda$ in the E plane and 0.375λ in the H plane. (b) What are the beamwidths? (c) What is the directivity?



Solution

Taking $\delta = \lambda/5$ in the E plane, we have

$$L = \frac{a^2}{8\delta} = \frac{100\lambda}{8/5} = 62.5\lambda$$

the flare angle in the E plane

$$\theta_E = 2\tan^{-1}\frac{a}{2L} = 2\tan^{-1}\frac{10}{125} = 9.1^\circ$$

Taking $\delta = 3\lambda/8$ in the H plane we have

$$\theta_H = 2\cos^{-1}\frac{L}{L+\delta} = 2\cos^{-1}\frac{62.5}{62.5+0.375} = 12.52^\circ$$

H-plane aperture

$$a_H = 2L\tan\frac{\theta_H}{2} = 2\times 62.5\lambda\tan 6.26^\circ = 13.7\lambda$$

$$\text{HPBW } (E \text{ plane}) = \frac{56^\circ}{a_{E\lambda}} = \frac{56^\circ}{10} = 5.6^\circ$$

$$\text{HPBW } (H \text{ plane}) = \frac{67^\circ}{a_{H\lambda}} = \frac{67^\circ}{13.7} = 4.9^\circ$$

$$D \simeq 10 \log\left(\frac{7.5 A_p}{\lambda^2}\right) = 10 \log(7.5 \times 10 \times 13.7) = 30.1 \text{ dBi}$$

Module 5: Chapter 2

Antenna Types:

- Helical Antenna
- Helical Geometry
- Practical Design Considerations of Helical Antenna,
- Yagi-Uda array
- Parabola General Properties
- Log Periodic Antenna

Suggested Reading

Text 3:"Antennas and Wave Propagation"

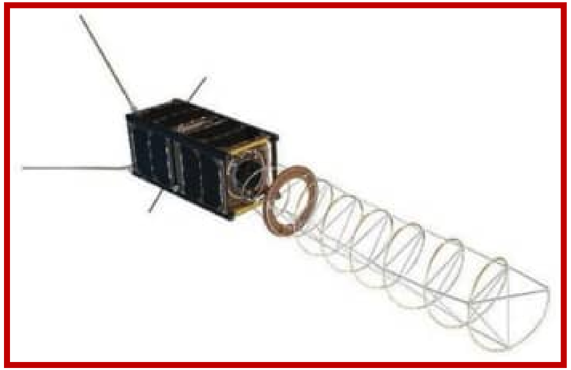
John D. Krauss, Ronald J Marhefka and Ahmad S Khan, 4th Special Indian Edition , McGraw- Hill Education Pvt. Ltd., 2010.

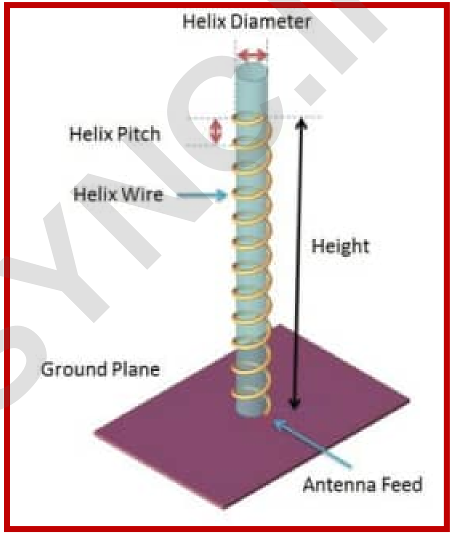
Text 3:

Sections: 8.3, 8.5, 8.8, 9.5, 11.7

Helical Antenna

A helical antenna is an antenna consisting of one or more conducting wires wound in the form of a helix.




Helix Diameter
Helix Pitch
Helix Wire
Height
Ground Plane
Antenna Feed

Introduction to Helical Antenna

The concept of helical antenna was first introduced by John D. Kraus in 1947. It is 3-dimentional structure of which linear and loop antennas are the special cases. It is basically a simple broadband VHF and UHF antenna which provides circular polarization. It consists of a thick copper wire wound in the form of a screw thread forming a helix. When the helical wire is unwounded on a flat surface it becomes a straight wire. The wire is wounded in such a way that it is wounded as if on a uniform cylinder. When it is observed from any end, the shape observed is circular. Thus the helix of a helical antenna combines three different geometric shapes like straight line, circle and a cylinder. Moreover it can be right handed or left handed.

Modes of Helical Antenna

Eventhough, a helix radiates in many modes, the modes of special interest are **normal mode** and **axial mode**. In the **axial mode**, the maximum radiation is along the helix axis under condition that the circumference of the helix is of the order of one wavelength as shown in Fig. 6.1.1 (b). In **normal mode**, the radiation is maximum along the broadside to the helix axis under condition that the circumference of the helix is smaller with respect to one wavelength. The radiation mode is illustrated in the Fig. 6.1.1 (a). When dimensions of the helix circumference exceed those required for normal mode and axial mode, a multilobed pattern as shown in the Fig. 6.1.1 (c) is observed. Such higher order radiation mode is called **conical mode**.

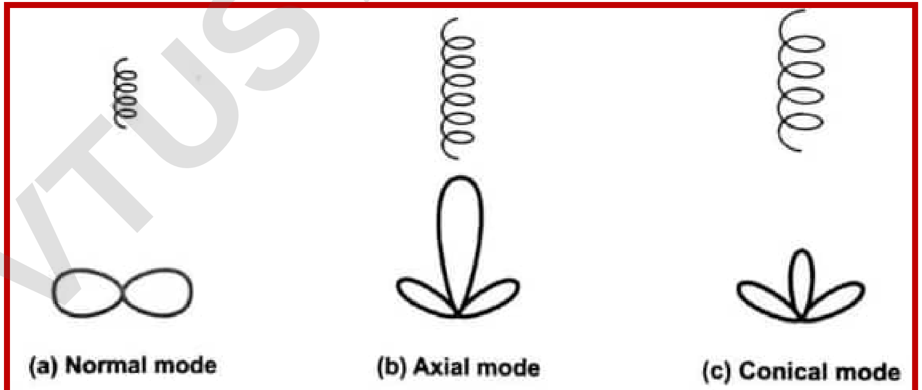
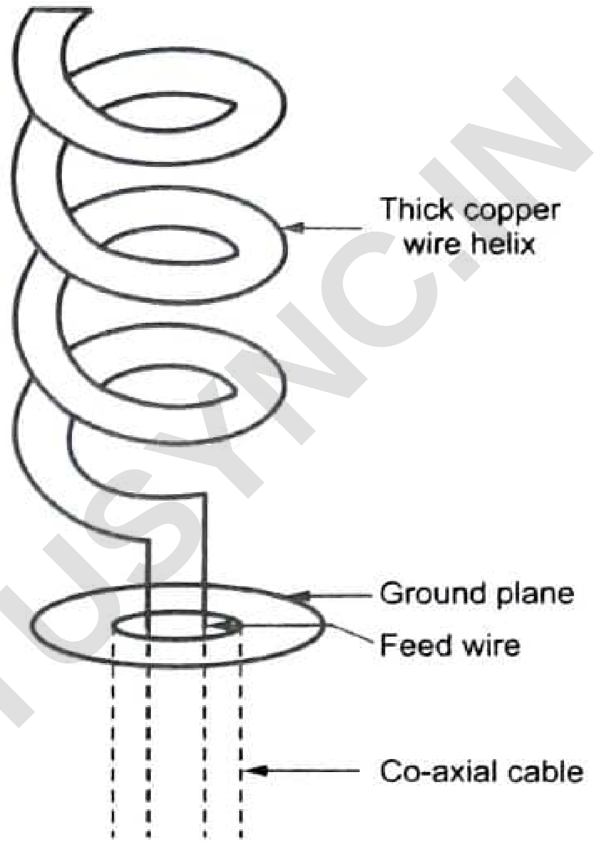


Fig. 6.1.1 Different modes of helical antenna

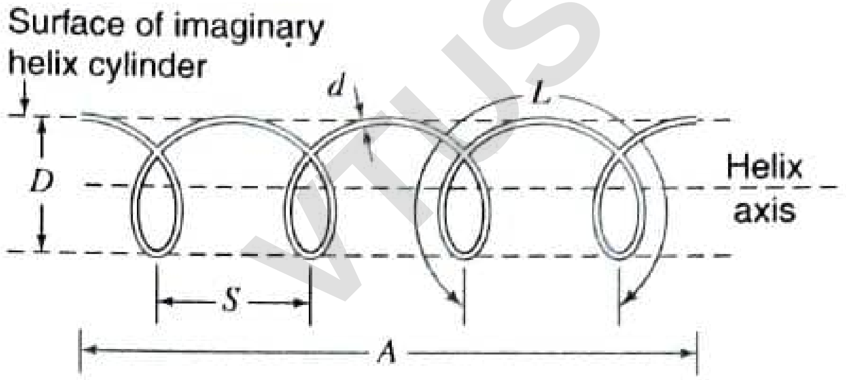
Helical Antenna Structure and Helical Geometry



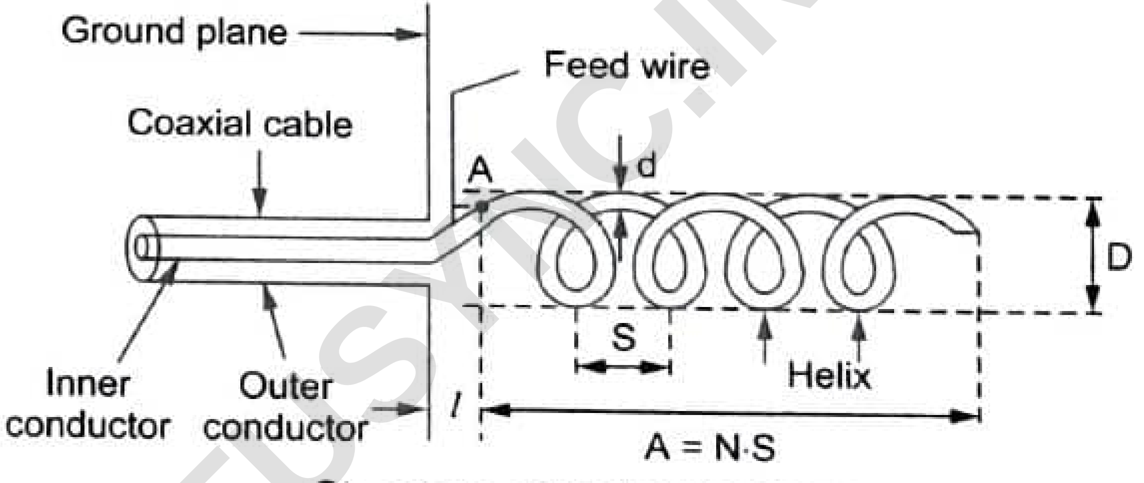
Helical antenna

Helical Geometry

The helix is a basic three-dimensional geometric form. A helical wire on a uniform cylinder becomes a straight wire when unwound by rolling the cylinder on a flat surface. Viewed end-on, a helix projects as a circle. Thus, a helix combines the geometric forms of a straight line, a circle and a cylinder. In addition a helix has handedness; it can be either left- or right-handed.



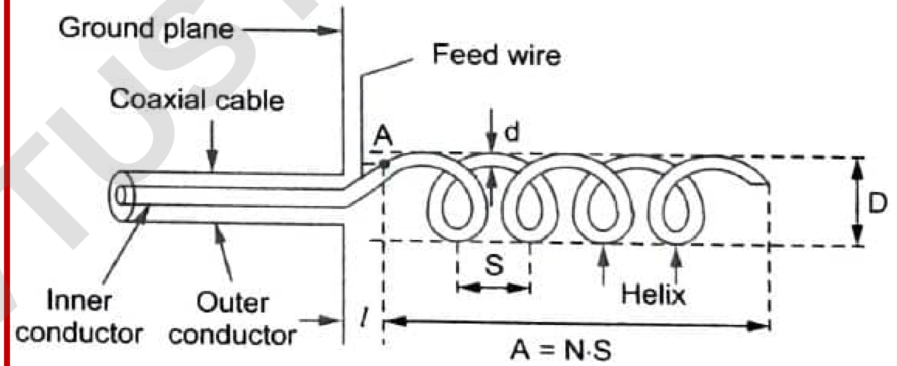
D = diameter of helix (center to center)
C = circumference of helix $= \pi D$
S = spacing between turns (center to center)
α = pitch angle $= \arctan S/\pi D$
L = length of 1 turn
n = number of turns
A = axial length $= nS$
d = diameter of helix conductor



Structure of helical antenna

The helix is fed by a coaxial cable. One end of the helix is connected to the centre or inner conductor of the cable and outer conductor is connected to the ground plane. The mode of the radiation of the antenna depends on the diameter of the helix i.e. D, the spacing between turn S which is measure between two centres of the turns. The circumference of the helix is denoted by C and it is equal to πD. Then the pitch angle α is given by,

$$\alpha = \tan^{-1}\left(\frac{S}{\pi D}\right)$$



Structure of helical antenna

The number of turns are denoted by N. Then the axial length A = N × S. Length of one complete turn is denoted by L. The spacing of the helix from the ground plane is denoted by l.

For N turns of the helix, the total length of the antenna is equal to N·S. While the circumference equals to πD. If we unroll one turn of helix on a plane surface, then the circumference, spacing between turns (S), turn length (L) and pitch angle (α) can be related to each other through the triangle terminology as shown in the Fig. 6.2.3.

Then we can write,

$$L = \sqrt{S^2 + C^2} = \sqrt{S^2 + (\pi D)^2}$$

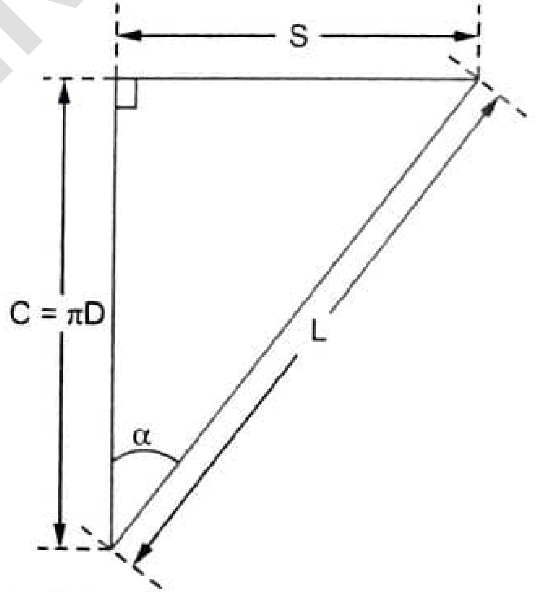


Fig. 6.2.3 Triangle terminology for one turn of helix unrolled on a plane surface

The **pitch angle** is defined as the angle between a line tangential to the helix wire and the plane normal to the axis of helix. The pitch angle then can be expressed as,

$$\tan\alpha = \frac{S}{C} = \frac{S}{\pi D}$$

$$\therefore \quad \boxed{\alpha = \tan^{-1}\left(\frac{S}{\pi D}\right)}$$

In general, a helical antenna can radiate in many modes. But the most important modes of radiation are as follows.

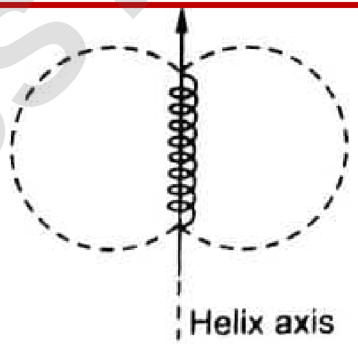
i) Normal or perpendicular mode of radiation.

ii) Axial or End fire or Bean mode of radiation.

Normal or Perpendicular Mode of Radiation

In this mode of helical antenna, the radiation is maximum in broadway direction i.e. Normal or perpendicular to the axis of the helix, hence the mode is called **normal or perpendicular mode of radiation**. The radiation in the direction normal to the helix axis is circularly polarized wave.

This mode of radiation can be obtained if the helix dimensions are made very small as compared with radiation is $N \cdot S << \lambda$. But with this mode of radiation, the bandwidth of antenna becomes narrow and the radiation efficiency also becomes very less.



Normal mode radiation pattern with circular polarization for helical antenna

Consider a helix in spherical co-ordinate system as shown in the Fig. 6.2.5. Considering that the helical antenna is made up of number of small loops and short dipoles arranged in series such that loop diameter equal to helix diameter D and length of the short dipole equal to the spacing between two helix S.

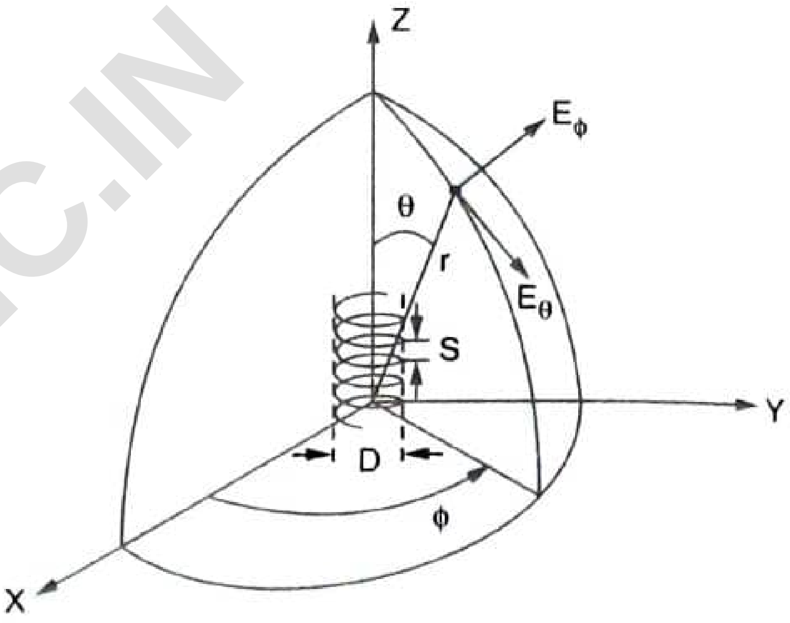


Fig. 6.2.5 Helical antenna with spherical co-ordinates

The far field of a small loop is given by,

$$E_{\phi} = \frac{120\pi^2 [I] \sin\theta}{r} \cdot \frac{A}{\lambda^2} \quad \ldots(6.2.1)$$

where $[I]$ = Retarded current, A

r = Distance at a point, m

A = Area of loop $= \pi\left(\frac{D}{2}\right)^2 = \frac{\pi D^2}{4}\ m^2$

λ = Wavelength in m.

Similarly the far field of the short dipole is given by,

$$E_{\theta} = j\frac{60\pi [I] \sin\theta}{r} \cdot \frac{S}{\lambda} \quad \ldots(6.2.2)$$

where $S = dL$ = Length of the dipole

From above equations, it is clear that two fields are in phase quadrature (phase of 90°). To get axial ratio (AR) of the elliptical polarization, we should take ratio of magnitudes of the field due to short dipole to that due to the loop. Hence the axial ratio is given by,

$$AR = \frac{|E_\theta|}{|E_\phi|} = \frac{\left| j\dfrac{60\pi[I]\sin\theta}{r} \cdot \dfrac{S}{\lambda} \right|}{\left| j\dfrac{120\pi^2[I]\sin\theta}{r} \cdot \dfrac{A}{\lambda^2} \right|} = \frac{S\lambda}{2\pi A}$$

Substituting value of A as $\frac{\pi D^2}{4}$, we get,

$$AR = \frac{S\lambda}{2\pi\left(\dfrac{\pi D^2}{4}\right)} = \frac{4S\lambda}{2\pi^2 D^2} \quad \therefore \quad \boxed{AR = \frac{2S\lambda}{\pi^2 D^2}}$$

Now depending on values of AR, we get three conditions.

Condition 1 : When AR = 0, the elliptical polarization becomes linear horizontal polarization,

Condition 2 : When AR = ∞, the elliptical polarization becomes linear vertical polarization,

Condition 3 : When AR = 1, the elliptical polarization becomes circular polarization.

Thus the condition for the circular polarization is given by,

$$AR = 1 = \frac{|E_\theta|}{|E_\phi|} = \frac{2S\lambda}{\pi^2 D^2}$$

i.e. $|E_\theta| = |E_\phi|$

Hence, we can write,

$$2S\lambda = \pi^2 D^2$$

i.e.

$$\boxed{S = \frac{\pi^2 D^2}{2\lambda} = \frac{C^2}{2\lambda}} \qquad \ldots(6.2.3)$$

where, C = Circumference = πD

Hence the pitch angle for the circular polarization is given by,

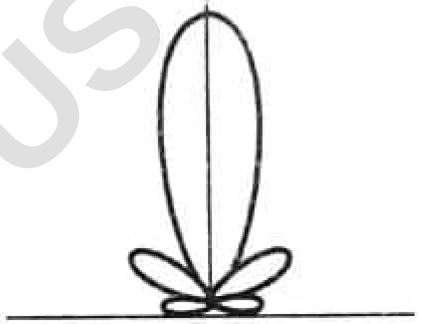
$$\alpha = \tan^{-1}\left(\frac{S}{\pi D}\right) = \tan^{-1}\left(\frac{\frac{\pi^2 D^2}{2\lambda}}{\pi D}\right)$$

$$\therefore \quad \boxed{\alpha = \tan^{-1}\left(\frac{\pi D}{2\pi}\right) = \tan^{-1}\left(\frac{C}{2\lambda}\right)} \qquad \ldots(6.2.4)$$

where, C = Circumference = πD

Axial or End Fire Mode of Radiation

The helical antenna radiating field maximum in the end fire direction i.e. along the axis of the helix is called **axial mode or end fire mode helical antenna.** With the axial mode radiation, the polarization of the wave is either circular or nearly circular. The main difference in the radiation pattern of the normal mode and axial mode is that in the axial mode, radiation is maximum along the helix axis while in normal mode, radiation is maximum in the direction normal to the helix axis.



Radiation pattern of helical antenna in axial mode

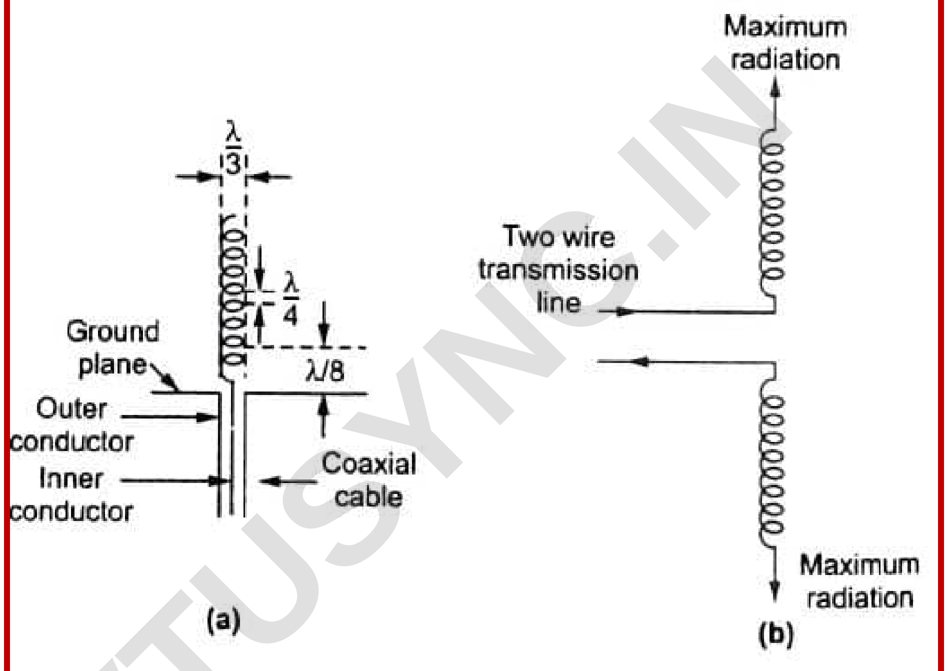


Fig. 6.2.7 Helical antenna in axial mode

For the axial mode, the radiation pitch angle α varies from 12° to 18°. The optimum pitch angle is 14°.

$$R = \frac{140C}{\lambda} \quad \ldots(6.2.5)$$

In the axial mode, the antenna gain and beam width both depend on the length of helix i.e. $N \cdot S$. The beamwidth between half power points is given by,

$$HPBW = \frac{52}{C}\sqrt{\frac{\lambda^3}{N \cdot S}} \text{ degrees} \quad \ldots(6.2.6)$$

Similarly the beam width between first nulls is given by,

$$BWFN = \frac{115}{C}\sqrt{\frac{\lambda^3}{N \cdot S}} \text{ degrees}$$

The maximum directive gain in the axial mode is given by,

$$G_D = \frac{15 N S C^2}{\lambda^3} \quad \ldots(6.2.7)$$

The axial ratio (AR) is given by,

$$AR = 1 + \frac{1}{2N} \quad \ldots(6.2.8)$$

Salient Features of Helical Antenna

The salient features of helical antenna are as follows :

i) Helical antenna is used for circular polarization.

ii) The helical antenna is used most widely in VHF and UHF bands.

iii) The axial mode of helical antenna is most widely used.

iv) The anenna in axial mode has larger bandwidth but in normal mode bandwidth and efficiency both are small.

v) Its construction is simple and directivity is higher.

Example 6.2.1 *Calculate directivity of 20 turn helix with $\alpha = 12°$ and circumference equal to one wavelength.*

Solution : The directivity of helical antenna is given by,

$$G_{D_{max}} = \frac{15 N S C^2}{\lambda^3}$$

where, N = Turns of helix = 20, C = Circumference = λ, S = Spacing between turns.

By property,

$$\tan\alpha = \frac{S}{C} \quad \text{i.e.} \quad S = C \cdot \tan\alpha = \lambda(\tan 12°) = \mathbf{0.2125\ \lambda}$$

$$\therefore \quad G_{D_{max}} = \frac{15 \times 20 \times 0.2125\lambda \times (\lambda^2)}{\lambda^3} = \mathbf{63.75}$$

Example 6.2.2 *For a 20 turn helical antenna operating at 3 GHz with circumference of 10 cm and spacing between the turns 0.3 λ is operating at 3 GHz. Calculate the directivity and half power beamwidth.*

Solution : $N = 20$, $C = 10\text{ cm} = 10\times10^{-2}\text{m}$, $S = 0.3\,\lambda$, $f = 3\text{ GHz}$.

i) The wavelength is given by,

$$\lambda = \frac{c}{f} = \frac{3\times10^{8}}{3\times10^{9}} = 0.1\text{ m}$$

ii) The directivity is given by,

$$G_{D\max} = \frac{15NSC^2}{\lambda^3} = \frac{15\times20\times0.3\times0.1\times10\times10^{-2}}{(0.1)^3} = 900$$

iii) Half power beam width is given by,

$$HPBW = \frac{52}{C}\sqrt{\frac{\lambda^3}{NS}} = \frac{52}{10\times10^{-2}}\sqrt{\frac{(0.1)^3}{20\times0.3\times0.1}} = 21.23^\circ$$

Example 6.2.4 *A 16-turn helical beam antenna has a circumference of λ, and turn spacing of $\lambda/4$. What is i) HPBW ii) axial ratio iii) gain and iv) power pattern ?*

Given : N = Number of turns = 16, S = Spacing = $\lambda / 4$, C = Circumference = λ

i) Beamwidth between half power points is given by,

$$\text{HPBW} = \frac{52}{C}\sqrt{\frac{\lambda^3}{NS}} = \frac{52}{\lambda}\sqrt{\frac{\lambda^3}{16(\lambda/4)}} = \frac{52}{\lambda}\sqrt{\frac{\lambda^2}{4}} = 26^\circ$$

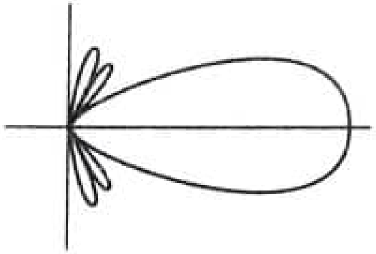
ii) Axial ratio is given by,

$$AR = 1 + \frac{1}{2N} = 1 + \frac{1}{2(16)} = 1.03125$$

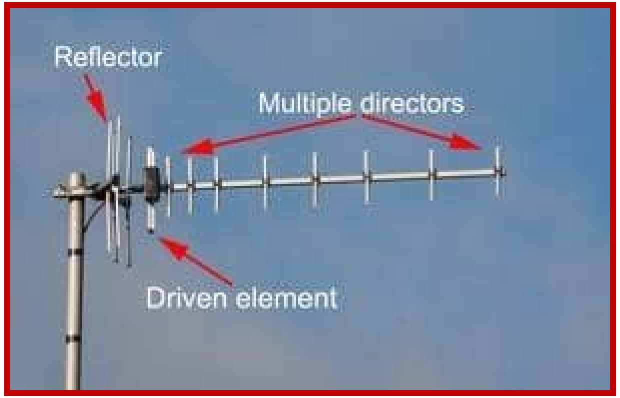
iii) Directive gain is given by,

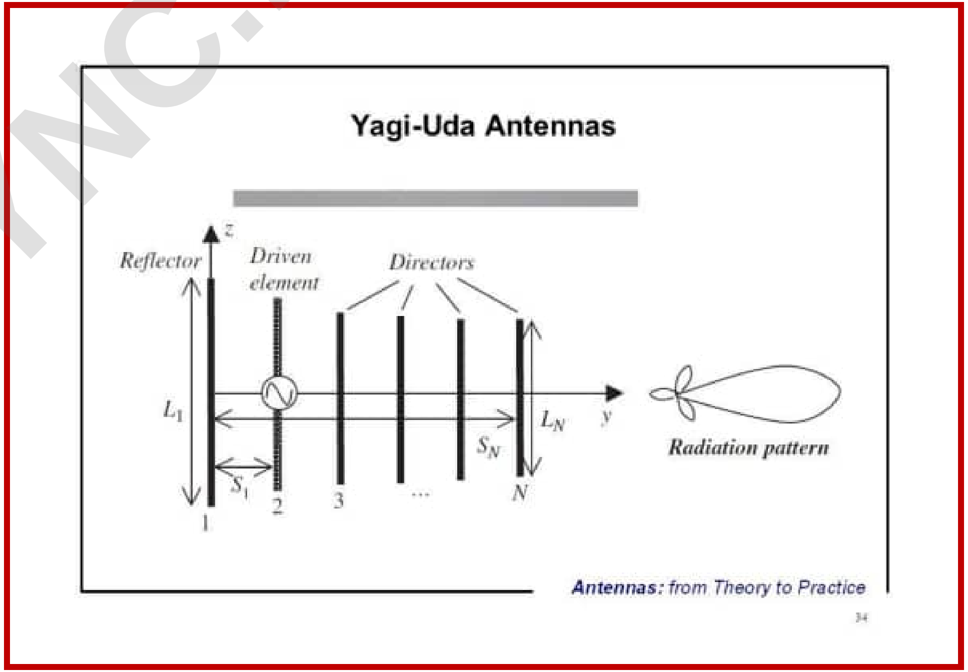
$$D = \frac{15NSC^2}{\lambda^3} = \frac{15 \times 16 \times \left(\frac{\lambda}{4}\right)(\lambda)^2}{\lambda^3} \; 60$$

iv) Power pattern is as shown below.



Yagi-Uda Antenna

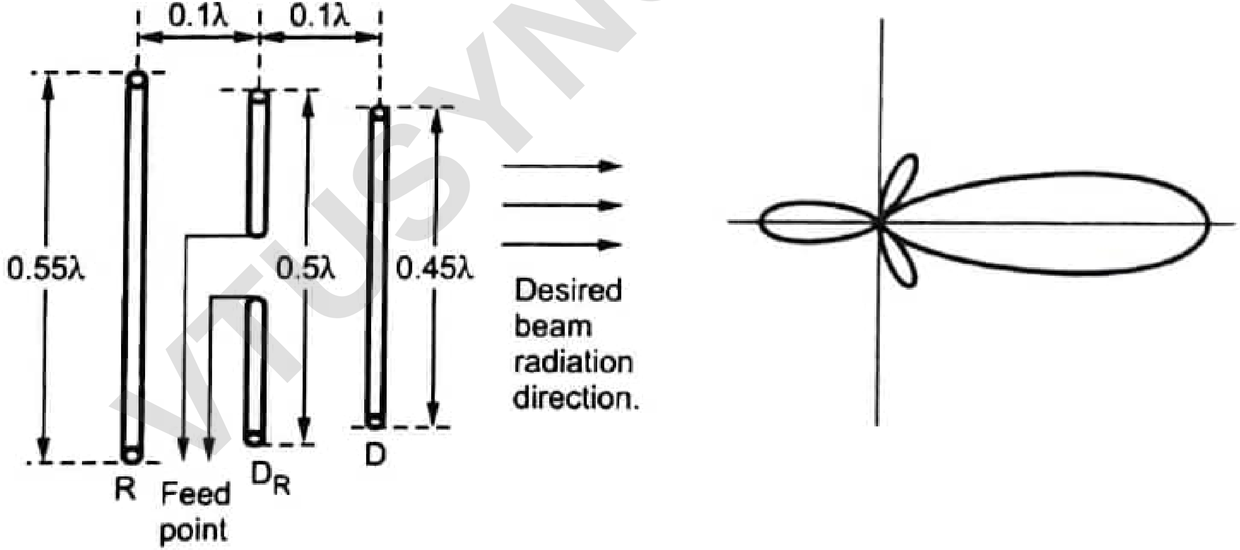

Reflector
Multiple directors
Driven element


Yagi-Uda Antennas
Reflector
Driven element
Directors
z
y
L_1
L_N
S_1
S_N
1
2
3
...
N
Radiation pattern
Antennas: from Theory to Practice
34

Introduction

Yagi-Uda arrays or Yagi-Uda antennas are high gain antennas. The antenna was first invented by a Japanese **Prof. S. Uda** in early 1940's. Afterwards it was described in English by **Prof. H. Yagi**. As the description was in english, it was read worldwide and the antenna became popular. Hence the antenna name Yagi-Uda antenna was given after Prof. S. Uda and Prof. H. Yagi. Prof. Uda performed several experiments. He measured gains and patterns with single parasitic reflector, single parasitic director and with a reflector and as many as 30 directors. He found that highest gain is possible with the reflector of length equal to $\frac{\lambda}{2}$ located at a distance $\frac{\lambda}{4}$ from the driven element, along with director of length approximately 10 % less than $\frac{\lambda}{2}$ located at a distance $\frac{\lambda}{3}$ from the driven element.

A basic Yagi-Uda antenna consists a driven element, one reflector and one or more directors. Basically it is an array of one driven element and one of more parasitic elements. The driven element is a resonant half wave dipole made of a metallic rod. The parasitic elements which are continuous are arranged parallel to the driven elements and at the same line of sight. All the elements are placed parallel to each other and close to each other as shown in the Fig. 6.4.1.



The parasitic element receive excitation through the induced e.m.f. as current flows in the driven element. The phase and amplitude of the currents through the parasitic elements mainly depends on the length of the elements and spacing between the elements. To vary reactance of any element, the dimensions of the elements are readjusted. Generally the spacing between the driven and the parasitic elements is kept nearly 0.1 λ to 0.15 λ. A Yagi-Uda antenna uses both the reflector (R) and the director (D) elements in same antenna. The element at the back side of the driven element is the reflector. It is of the larger length compared with remaining elements. The element infront of the driven element is the director which is of lowest length in all the three elements. The lengths of the different elements can be obtained by using following formula :

$$\text{Reflector length} = \frac{152}{f\,(\text{MHz})}\ \text{meter}$$

$$\text{Driven element length} = \frac{143}{f\,(\text{MHz})}\ \text{meter}$$

$$\text{Director length} = \frac{137}{f\,(\text{MHz})}\ \text{meter}$$

Yagi-Uda Antenna Calculations

To design Yagi-Uda antenna properly it is important to know the wavelength of the electromagnetic wave. The wavelength of the electromagnetic wave can be obtained by using relationship given as,

$$\lambda = \frac{c}{f} \qquad \text{... (6.4.1)}$$

where, c = Velocity of the light = 3×10^8 m/sec

f = Frequency of electromagnetic wave

We know that length of the dipole is given as,

$$L = \frac{\lambda}{2} \qquad \text{... (6.4.2)}$$

Putting value of λ in equation (6.4.2), we get,

$$L = \frac{c/f}{2} = \frac{3 \times 10^8 / f\,(\text{MHz})}{2} = \frac{150}{f} \text{ meter}$$

$$\therefore \quad \boxed{L = \frac{150}{f\,(\text{MHz})} \text{ meter}} \qquad \text{... (6.4.3)}$$

Due to electrical characteristics of the antenna material it is found that the antenna elements should be 5 % to 7 % shorter in practice than those obtained using exact formula. Hence we can rewrite expression for L as,

$$\boxed{\text{For dipole : } L = \frac{143}{f\,(\text{MHz})} \text{ meter}} \quad \dots (6.4.4)$$

The lengths of reflector and first director are given by expression given below.

$$\boxed{\text{For Reflector : } L = \frac{152}{f\,(\text{MHz})} \text{ meter}} \quad \dots (6.4.5)$$

And,

$$\boxed{\text{For } 1^{st} \text{ Director } D_1 : L = \frac{137}{f\,(\text{MHz})} \text{ meter}} \quad \dots (6.4.6)$$

The spacing between reflector R and dipole D_R is given by,

$$\text{Spacing between R and } D_R = 0.25\,\lambda = \frac{75}{f\,(\text{MHz})} \text{ meter} \qquad \dots (6.4.7)$$

Similarly the spacing between dipole D_R and director D_1 is given by,

$$\text{Spacing between } D_R \text{ and } D = 0.13\,\lambda = \frac{40}{f\,(\text{MHz})} \text{ meter} \qquad \dots (6.4.8)$$

The spacing between two directors D_1 and D_2 is given by,

$$\text{Spacing between } D_1 \text{ and } D_2 = \frac{38}{f\,(\text{MHz})} \text{ meter} \qquad \dots (6.4.9)$$

Key Point *When more number of directors are required then the length of the subsequent director shortens by 2.5 %.*

Example 6.4.1 *Calculate the lengths of the dipole, reflector, director 1, director 2 and director 3 of the Yagi-Uda antenna alongwith the spacings between the elements for channel 5 (VHF-III).*

Solution : The band frequency for channel 5 (VHF-III) is from 174 MHz to 181 MHz.

For antenna calculations we can use centre frequency. The centre frequency f can be calculated as,

$$f = \frac{(174+181)\,10^6}{2} = 177.5\times10^6 = 177.5 \text{ MHz}$$

$$\text{Length of dipole } D_R = L = \frac{143}{f\,(\text{MHz})} = \frac{143}{177.5} = \mathbf{0.80\ m}$$

$$\text{Length of reflector } R = \frac{152}{f\,(\text{MHz})} = \frac{152}{177.5} = 0.856\ \text{m} \approx \mathbf{0.86\ m}$$

$$\text{Length of director 1 (i.e. } D_1) = \frac{137}{f\,(\text{MHz})} = \frac{137}{177.5} = \mathbf{0.77\ m}$$

$$\text{Length of director 2 (i.e. } D_2) = \frac{133}{f\,(\text{MHz})} = \frac{133}{177.5} = 0.749\ \text{m} \approx \mathbf{0.75\ m}$$

$$\text{Length of director 3 (i.e. } D_3) = \frac{130}{f\,(\text{MHz})} = \frac{130}{177.5} = \mathbf{0.73\ m}$$

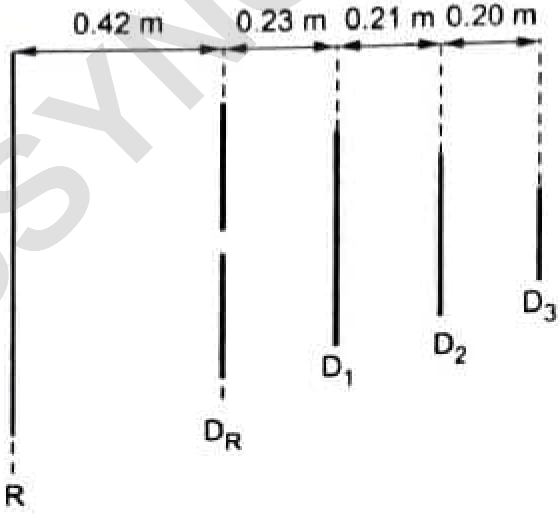
$$\text{Spacing between reflector and dipole} = \frac{75}{f\,(\text{MHz})} = \frac{75}{177.5} = \mathbf{0.42\ m}$$

$$\text{Spacing between dipole and director 1} = \frac{40}{f\,(\text{MHz})} = \frac{40}{177.5} = 0.225\ \text{m} \approx \mathbf{0.23\ m}$$

Spacing between director 1 and director 2 = $\frac{38}{f\,(MHz)} = \frac{38}{177.5} = 0.21$ m

Spacing between director 2 and director 3 = $\frac{36}{f\,(MHz)} = \frac{36}{177.5} = 0.20$ m

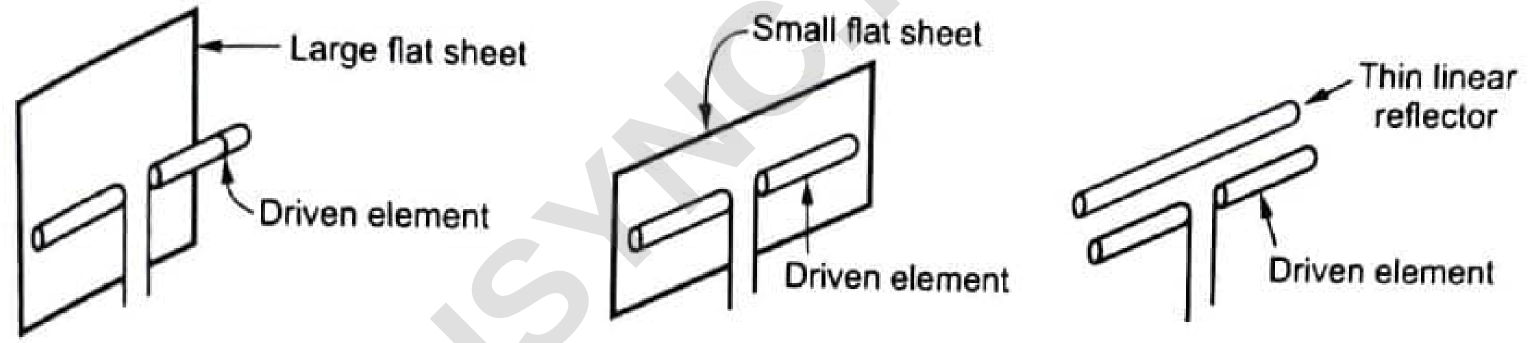
The dimensional sketch of antenna is as shown in the Fig. 6.4.4.



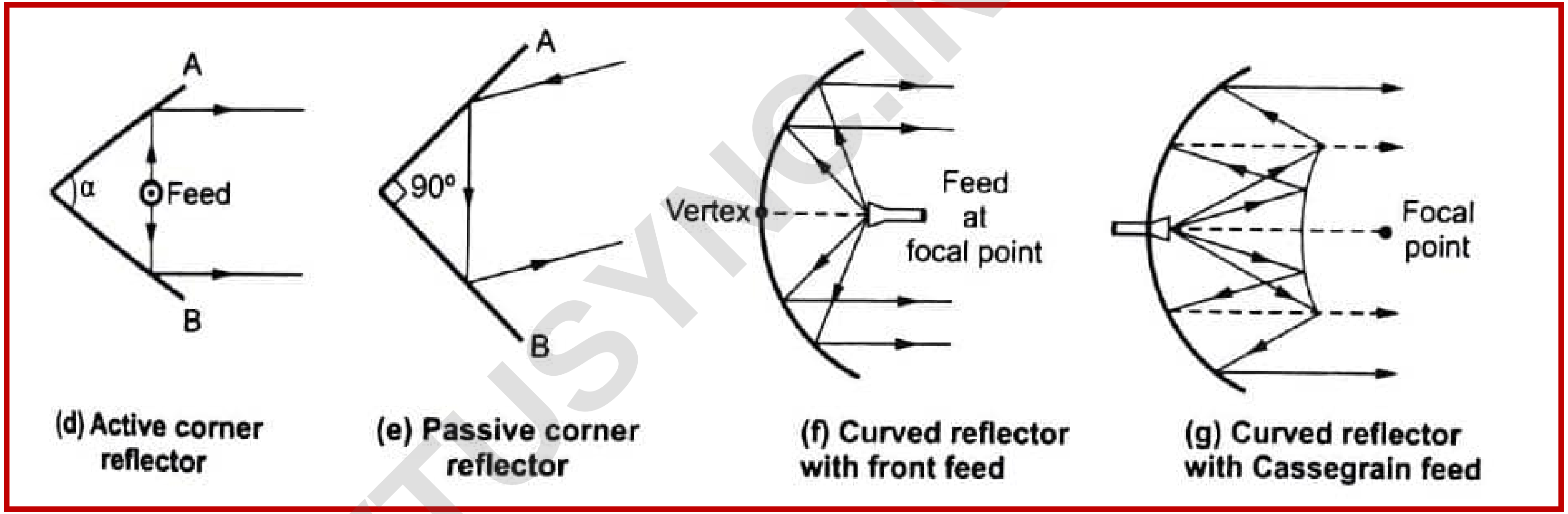
Reflector Antennas

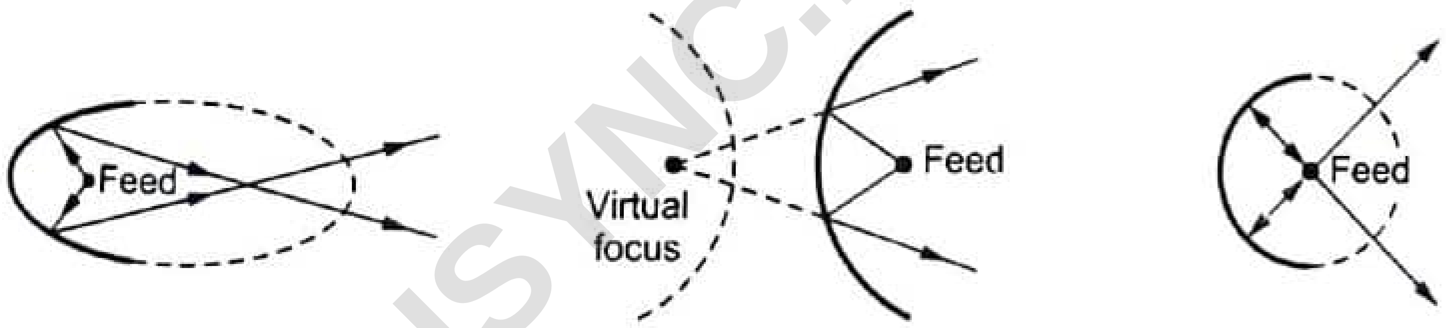
- The reflector antennas are most important in microwave radiation applications.
- At microwave frequencies the physical size of the high gain antenna becomes so small that practically any suitable shaped reflector can produce desired directivity.
- In reflector antenna, another antenna is required to excite it.
- Hence the antenna such as dipole, horn, slot which excites the reflector antenna is called **primary antenna**, while the reflector antenna is called **secondary antenna**.
- In general, reflector antenna can be represented in any geometrical configuration, but the most commonly used shapes are plane reflector, corner reflector and curved or parabolic reflectors.
- Using reflectors, the radiation pattern of a radiating antenna can be modified.

Various possible configurations of reflector antennas:



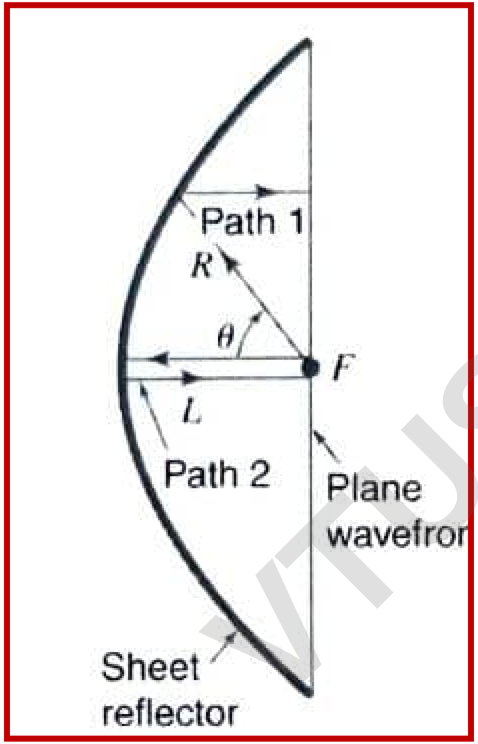
(a) Large flat sheet (b) Small flat sheet (c) Thin linear reflector antenna


A
B
α
Feed
(d) Active corner reflector
A
B
90°
(e) Passive corner reflector
Vertex
Feed at focal point
(f) Curved reflector with front feed
Focal point
(g) Curved reflector with Cassegrain feed



(h) Elliptical reflector (i) Hyperbolic reflector (j) Circular reflector

Parabolic reflectors



$$2L = R(1 + \cos\theta)$$

and

$$R = \frac{2L}{1 + \cos\theta}$$

Thank you